HAST-LNC: Hybrid Audio Spectrogram Transformer Embedding with Lightweight Neural Classifier for Infant Cry Analysis

Abstract

Infant cry analysis is critical in determining the basic infant needs including hunger, pain, discomfort and attention need. Manual interpretation of infant cries is, however, subjective, and often unreliable because of different variety of cries, recording conditions and background experience of the caregiver. To overcome these shortcomings, this study presents an automated infant cry recognition system called HAST-LNC. The proposed system is based on the HAST-LNC (Hybrid Audio Spectrogram Transformer Embedding with Lightweight Neural Classification) system of infant cry analysis. Under this method, audio signals of infant cry are initially transformed into spectrogram representations to achieve the time-frequency properties of the cry. Processing on these spectrograms is done with a pretrained Audio Spectrogram Transformer (AST) model which extracts high level acoustic embeddings through selfattention mechanisms that can learn more complex contextual representations of audio data. The AST model produces fixed length feature vectors of 768, which is used to represent acoustic patterns of infant cries. These embeddings are subsequently fed to lightweight classification models such as Logistic Regression and Multi-Layer Perceptron (MLP) and the underlying cause of the cry is predicted. Logistic Regression is applied as a reference probabilistic classifier to give estimated class probabilities, and the MLP classifier is trained to learn nonlinear decision-boundaries with hidden layers using ReLU activation functions and Softmax output functions to predict more than two classes. These learned embeddings are then classified through a baseline model of Logistic Regression and the main classifier of MLP. The experimental result shows that overall accuracy of the MLP classifier is 89.66 % which is higher than the baseline of the Logistic Regression which had an accuracy of 70.65%. Moreover, the proposed model enhances macro-average recall (0.33) and macro-average F1-score (0.30), which means that there is a more equal performance between the cry classes. The ASTMLP architecture is able to attain a high detection performance at a low level of computational complexity as a result of combining transformer-based representation learning with a computational-efficient neural classifier. In general, the hybrid feature extraction lightweight neural classification that excludes the need to use complex computations and offers high predictive power. The HASTLNC framework is therefore an efficient and scalable mechanism of automated infant cry analysis and thus it can be used in real-time smart monitoring systems and healthcare-support applications.

Keywords Deep Learning, Audio Spectrogram Transformer, Infant Cry Analysis, Machine Learning, Logistic Regression, Multi-Layer Perceptron, General Need Classification.

1. Introduction

The study of infant cry has developed into an important area of interest in the pediatric care and first-year infant development since infant crying has been the major method through which infants express their physical and emotional conditions (hunger, discomfort, pain etc.), and sleepiness [1]. Proper understanding of these vocal signs makes caregivers and health care professionals provide timely responses, which helps in enhancing the infant health and minimizing the stress of caregivers [2].

Recently machine learning methods have shown encouraging results on automated infant cry classification especially by using handcrafted acoustic features like Mel-Frequency Cepstral Coefficients (MFCCs) as well as spectrogram-based representations with standard classifiers [3].

However, in spite of these developments, infant cry signals are extremely non-stationary, multidimensional and affected by a range of physiological and environmental factors and thus are a very difficult signal to analyze reliably [4]. Traditional caregiverbased feature interpretation and handcrafted methods of feature are not usually robust and generalizable [5]. Although deep learning systems such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) have been demonstrated to have more powerful representation-learning properties[6], they may be expensive to calculate, and require inflexibility to diverse recording conditions, which makes them less useful in practice[7].

New opportunities in audio signal analysis have also been introduced by the recent transformer-based architectures as attention mechanisms [8] enable to capture long-range temporal responses and small spectral variations in a way that outperforms CNN- and RNN-based systems in audio classification [9].

Recommended in this paper, using these developments, is HAST-LNC: Hybrid Audio Spectrogram Transformer Embedding with Lightweight Neural Classification to Infant Cry Analysis. This structure has an Audio Spectrogram Transformer (AST) in

the pre-trained form to acquire the deep spectro-temporal embeddings of log-Mel spectrograms[10], and is directly preceded by a Logistic Regression as the baseline and Multi-Layer Perceptron (MLP) as the primary lightweight classifier. The synergistic application of transformer-based learning of features and effective neural classification ensures accuracy, robustness, and computational efficiency that are acceptable in real life and infant care environment which makes HAST-LNC an effective approach. [11].

1.1 Objective

This study would focus more on developing a valid and accurate infant cry classification regime with the view of predicting the overall infant needs. The aim of the paper is to have a better classification with the least compromise in terms of robustness to the different sample cries, and the different conditions of recording, by utilizing the transformer-based feature extraction approach by use of the Audio Spectrogram Transformer (AST) and machine learning classifiers[12]. The previous research has shown that the traditional classifiers along with the designed acoustic features can reach high levels of classification, however, the limitations are in the feature representation and generalization which can be surmounted with the help of transformer-based approaches.

1.2 Significance

The importance of the work is that it could benefit caregivers, parents, and healthcare providers with an automated and objective infant cry interpretation tool[13]. A precise system of infant cry classification would assist in reducing the stress of the caregivers, enhance the welfare of the babies and provide the caregivers with the opportunity to respond to the demands of the babies in a timely manner. Also, abnormal crying patterns should be detected early so that urgent clinical intervention can be delivered where appropriate, enhancing health outcomes[14]. Transformer-based audio representation has improved the generalization of the system to real-world situations and recording conditions, which has been identified as a major development in audio analysis studies.

1.3 Innovations in the Work

1.3a AST-Based Cry Feature Representation.

In this paper, the Audio Spectrogram Transformer (AST) is used to process and extract features of infant crying, which allows it to model the complex time-frequency features of cry sounds[15]. In contrast to the conventional handcrafted acoustic features, AST is learned to extract rich and discriminative features using spectrogram as input, long-range temporal features and fine spectral details.

1.3b Hybrid Classification Using Logistic Regression and MLP

The feature embeddings derived with AST are categorized with the help of a Logistic Regression as well as Multi-layers Perceptron (MLP) framework[16]. The hybrid assessment system facilitates a comparative study of linear and non-linear classification methods, which offers information about the interpretability and performance trade-off of the model, which is significant to real-world applications where both accuracy and computational efficiency is relevant[17].

1.3c End-to-End General Infant Need Prediction Framework

The suggested system combines audio preprocessing, feature extraction with the use of transformers, and machine learning classification into an integrated end-to-end pipeline[18]. This end-to-end design enhances strength and scalability to varied infant cry datasets, overcoming shortcomings of a systematic analysis of infant cry analytics, including scalability and dataset variations reported.

1.4 Ethical Considerations

A framework that is both ethically and practically deployed is provided. It is based solely on audio recordings of noninvasive cry and does not presuppose physical contact or intrusive sensory equipment[19]. This method maintains the comfort and privacy of infants, which can be utilized morally at home and in clinical settings.

2 Literature Survey

The infants largely use cries to convey their physiological and emotional needs such as hunger, discomfort, pain, or fatigue. The cry sounds cannot be given verbally and it is thus the responsibility of the caregivers to decode them so that they get to know how to react [20]. However, the semi-subjective nature of the process of manual interpretation is prone to regard the experience of caregivers, the sound of the surrounding environment, and infantal variations. The erroneous interpretation may delay the proper

treatment, which may affect the health of newborns and increase the degree of stress among caregivers, which justifies the need of the automated and objective systems of crying analysis [21]. In signal analysis terms, the infant cries are non-stationary and complex audio signals of enormous alterations in pitch, intensity, and spectral composition. The classical procedures involving the use of hand-made acoustic properties were found to be low resilience with regards to data sets and recording settings. The recent advancements in deep learning and transformer-based audio models permit a greater quantity of trustworthy extraction of informative illustrations of cry codes [22]. Therefore, the construction of good automated systems that entail good feature extraction and useful classification models has become an important area of research in the infant cry analysis process. In the research that has been carried out in the past about the classification of infant cry, the representations of signal and machine learning structure have been rich variety in terms of improving the accuracy of prediction. The first methods, which are founded on acoustic properties, produced acoustic MFCCs, acoustic pitch, acoustic entropy and spectral information, and submitted to classical classifiers such as Gaussian Mixture Model (GMM), Hidden Markov Model (HMM), k-Nearest Neighbors (k-NN), Support Vector Model (SVM) and Probabilistic Neural Networks (PNN)[23]. These models have also performed classification based on learned statistical distributions or decision boundaries by which the categories of state cry are based on learned acoustic patterns. Although these methods proved that it could be automatically recognized that cry existed, the discriminative ability of manually constructed feature was restricted to its recognition capability [24]. Later research has shifted to deep learning models which are automatically trained on representations of time - frequency representations like spectrograms. CNNs were quite popular to learn local spectral structures but hybrid networks such as CNN-RNN models and CRNN models learned time related modeling to learn how patterns of cry is changed with time [25]. The modeling process predicts the requirements of the infants having the aid of the learning hierarchical map properties of the input spectrogram and predicting the output classes through softmax based probability estimation. The rankings like Dunstan Baby Language and Donate-a-Cry experimental results have verified the deep buildings to be far better than more familiar ones especially where there are numerous classes. However, despite these, the problem of variability in data, small data samples, and extrapolation to other recording conditions, and more complex architectures, such as attention-based and transformer-based audio models, should be considered. This work found transformer-based acoustic representations and the Audio Spectrogram Transformer (AST) to analyze each infant cry audio file into a 768-dimensional fixed length embedding [26]. During the feature preparation, only the valid embeddings of the desired dimension were retained, and the corrupt or incompatible files were not subjected to any feature preparation to ensure consistency in the dataset. These were then trained on machine learning classifiers to obtain the feature matrix and label set [27].

To begin with, there was the use of the baseline model, the Logistic Regression to determine the separability of the AST features in a linear decision space. The needs of the infants are predicted by the model estimating the likelihood of a class through the softmax function and finally classifies each sample under the most likely of the classes. A Multi-Layer Perceptron (MLP) classifier has also been suggested to improve the performance of classification even further and is a non-linear decision boundary learner that is being trained with multiple hidden layers with ReLU activations. The training of MLP was preceded by normalization of features using standard scaling so as to stabilize optimization and improve convergence.

The outcomes of the experiment, implemented with references to the values of accuracy and the comprehensive report of the classification leads to the assumption that the accuracy scores with the hashtag of AST embedding and Logistic Regression and MLP classification lead to the conclusion that the accuracy scores, when utilizing the audio representations of transformer, are extremely effective regarding the predictive performance of the activity on the infant cry classification [28]. Although feature extractors such as AST based on a transformer provide rich spectral-temporal feature representations of infant cry, the small and skewed data sets available in the area can lead to the low transfer between recording conditions and groups of infants. Moreover, AST embeddings should be homogeneously processed, and may suffer noise and machine variability. Even the conventional classifiers are not free of their problems: In Logistic Regression, the condition of linear classification separability may not be met in complex acoustic patterns and MLP classifiers, but are able to model non-linear dependencies, they must be tuned, scale features and sufficient training data to prevent overfitting in particular when a minority of cry class is undersampled. These limitations imply that the AST with classical classifiers is prospective, yet the existing limitation to viability in practical cry classification of infants in the real world must still be improved in the aspects of diversification of datasets, optimisation of the model, and balancing the classes [29].

The proposed research will provide a comprehensive and helpful system of infant cry classification that will enable bridging the gap between the strong deep image representations and the deficient classifiers. The idea underlying the proposed solution is the profound time-frequency variability of infant cries training in accordance with transformer-based Audio Spectrogram Transformer (AST) embeddings. It is grounded on the reality that they do not need the hand-crafted acoustic features or heavy end to end deep neural networks but they learn rich time frequency cries of the babies through the assistance of embeddings [30]. The embeddings are an otherwise small, discriminative, representation of cry signals, and thus can be effectively trained using small datasets. In a bid to ensure good performance and simultaneously sensitivity to interpretation, the system implies two complementary classifiers; the Logistic Regression and a Multi-layer Perceptron (MLP). The Logistic Regression can offer the lightweight and interpretable base of features used in ascertaining the linear separability of the AST features, unlike the MLP that introduces the non-linear

modeling feature needed to optimize the accuracy of prediction in the different types of cries [31]. The use of this study and some combination of transformer representations and functional machine learning models can offer a solution to the baby monitoring in a natural setting, requiring it to be balanced, scalable and deployable and to be reliable, efficient and flexible in computations and flexibility.

3 Methodology

3.1 System Overview

HAST-LNC: Hybrid Audio Spectrogram Transformer Embedding with Lightweight Neural Classification of Infant Cry Analysis is the suggested framework that is an automatic framework which is capable of identifying the most important need-related cries and can be further subdivided into hunger, pain, discomfort, attention. Noise minimization, silence elimination and amplitude normalization and resampling is the first phase of preprocessing the raw audio files to be able to achieve uniformity. The denoised signals are converted into log-Mel spectrograms to retain time-frequency characteristics and the outputs are fed into a trained Audio Spectrogram Transformer (AST) to generate deep spectro-temporal embeddings. These embeddings produce high level representations of the acoustic features of infant cries that are fixed [32].

The feature vectors obtained are divided into stratified training and testing sets and tested with two classifiers: The Logistic Regression as a baseline model and Multi-layer Perceptron (MLP) as a major lightweight nonlinear classifier. To enhance stability and convergence, feature normalization is done prior to training MLP. Accuracy, recall, mean squared error and confusion matrix are used to measure the performance of the system. HAST-LNC (AST-MLP) provides an accurate, interpretable, and real-time infant cry recognition solution using transformer-based embedding learning, which is computationally-efficient to run, and appropriate to use in the real world [37].

3.1.a Proposed Methodology

The framework suggested is the HAST-LNC (Hybrid Audio Spectrogram Transformer Embedding with Lightweight Neural Classification) that is used to analyse infant cry, due to its feature representation power and computational efficiency. Many of the traditional infant cry classification techniques are based on hand-crafted acoustic features and sophisticated deep learning architectures, which need a lot of feature engineering and large amounts of computation. In its turn, HAST-LNC is based on using pretrained transformer-based audio embeddings that are capable of automatically learning detailed temporal and spectral patterns of infant cry signals. This lowers reliance on manual features extraction and enhances the capability of the model in illustrating subtle acoustic differences in the categories of cry like hunger, discomfort, and tiredness. Also, compared to heavy deep neural networks, the system can be trained with lightweight classifiers and can be easily interpreted and more adaptable to real-time usage.

The operation principle of the HAST-LNC framework starts with the raw infant cry audio recordings converted into spectrogram representations that give a visual image of variations in frequencies over a period of time. The spectrograms are then subjected to get deep contextual embeddings by running them through the Audio Spectrogram Transformer (AST) model which summarizes the significant acoustic properties of the cry signal. The resulting embeddings form the input of lightweight classification models that are trained to find decision boundaries between the various cry categories. These transformer-based representations are analyzed by the classifier and the most likely cry class is predicted using learnt patterns. The HAST-LNC framework can support infant cry recognition with high accuracy, scalability and computational efficiency through the combination of transformer based feature extraction and efficient classification, which are useful in smart monitoring and healthcare-support systems.

Proposed Study: https://github.com/Deekshita-G/General_Needs_Classification

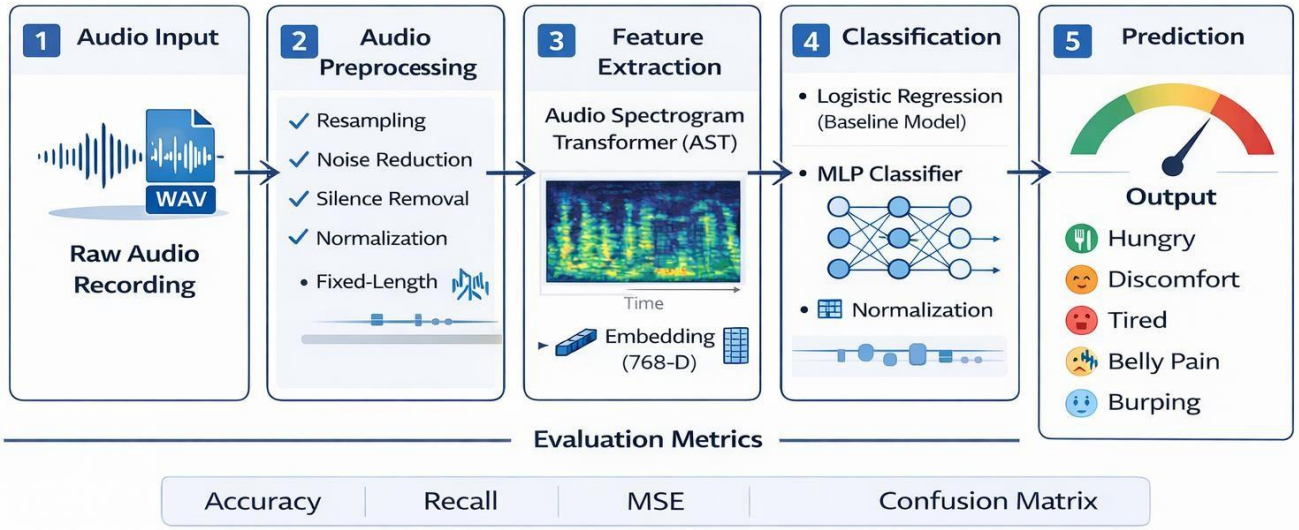


Fig 1. General cry type Multiclass Cry Classification Performance

3.2 Data Collection and Preprocessing

In this study, the dataset comprised the Donate-a-Cry infant audio corpus and it is summarized in Table 1. This dataset was 457 actual infant cry records based on user uploads, and thus, it is pertinent to the real-world home setting as opposed to controlled laboratory conditions. All audio samples are in WAV format and have lengths of about 6 to 7 seconds and the files have weights of 102 KB to 110 KB. As the recordings are not edited or screened, they also contain the natural fluctuations in background noise, recording apparatus and acoustics [33]. This diversity makes the data particularly, ultimately, handy in terms of the evaluation of the might and the generalization ability of machine learning models in a real-life context of infant cry classification. The methodology starts by the audio recordings of infant cry recordings that have been compiled upon sampled curated infant cry studies reported in Table 1 infant cry studies. These are the recordings which are taken under various acoustic conditions and are not fixed in duration, intensity and in the background sound. Such a variability implies that preprocessing is to be carefully done to make things consistent and reliable [10]. Preprocessing stage includes removal of noise, amplitude normalization, removal of silence and standardization of length which are widely applied in audio-based classification applications. The audio signal is processed before turning into frequency time representation as Mel spectrograms that provide a concise and perceptually decision representational form of cry signals that can be fed into deep learning models [32].

Table 1: Dataset Attributes and Description

Attribute	Details
Dataset Name	Infant Cry Audio Corpus (Donate-a-Cry)
Source	https://www.kaggle.com/datasets/bhoomikavalani/donateacrycorpusfeaturesdataset
Data Collection Method	User-uploaded infant cry recordings
Audio File Format	WAV file
Total Audio Files	457
Audio Nature	Real-world, unmodified, unchecked recordings
Audio Duration	6sec - 7sec
Audio File Size Range	102KB - 110KB

3.3 Audio Spectrogram Transformer (AST) Based Feature Extraction

In order to derive discriminative and robust representations on infant crystal signals, an already trained Audio Spectrogram Transformer (AST) model is used. AST takes inputs in the form of spectrograms and uses self-attention to model local and global dependencies along the time and frequency dimensions [34]. Compared to the traditional handcrafted feature extraction algorithms like MFCCs and pitch statistics, AST automatically learns rich features that represent finer acoustic differences that relate to various needs of infants. The transformer-based architecture allows the modelling of the long-range temporal dependence and complex spectral structures in both cases, which are crucial with regard to non-stationary audio signals like infant cries. The resulting AST embeddings create a more compact feature space that is rich in information and thus becomes input to the classification stage [6].

3.4 Classification Using Machine Learning and Neural Networks

General infant needs are classified with lightweight models by using the disaggregated AST embeddings. The use of Logistic Regression is based on the fact that it is simple, interpretable, and cheap to compute thus suitable in real-time use. Simultaneously, a Multi-Layer Perceptron (MLP) classifier is employed to learn non-linear features among the feature space that will allow better discrimination among the cry classes of infant [35]. The training of these classifiers is done on labelled AST feature embeddings to distinguish between the different infants cry classes. Transformer-based feature extraction combined with efficient classifiers provides the balance between classification accuracy and computational efficiency, which meets the practical deployment limitations.

3.5 Model Training and Optimization

The classifiers get to learn the best decision boundaries during training, using the feature representations obtained on the basis of AST [24]. The hyperparameters of learning rate, the count of hidden layers, the count of neurons, batch size, and the regularization terms are optimally adjusted to avoid overfitting and improve the generalization performance, Mathematically, training process is done by minimizing a classification loss function of the extracted feature space with the use of gradient-based optimization techniques [36]. The accuracy and consistency of prediction across the classes of infant cries are used as the metrics of model performance to guarantee the consistency of reliability in the identification of various types of infant cries in validation data.

3.6 Overall System Integration

The entire processing flow combines audio processing, time scale normalization, time-frequency conversion, feature extraction based on AST, and the classification using lightweight models into one pipeline, as depicted in Fig. 1. This systematic joint learning of transformer-based representation learning and efficient classifiers is more robust to cry variability and recording noise, and remains practicable in the real world [37]. The suggested structure is successful in responding to the troubles connected with the diversification of the infant cry and offers a trustworthy resolution to detect the overall infant requirements connected with hunger, discomfort, fatigue, and belly pain.

3.7 Model Parameters and Configuration

The major parameters employed to train the models and classify them can be summarized in Table 1, as well as the suggested values that were taken based on the previous works in audio classification and transformer-based learning [34]. The suggested values of parameters offer a good trade-off between model performance and accuracy, as well as the computer efficiency [38]. These values can be considered a good starting point and can be improved depending on the properties of data sets, validation performance, and the deployment conditions.

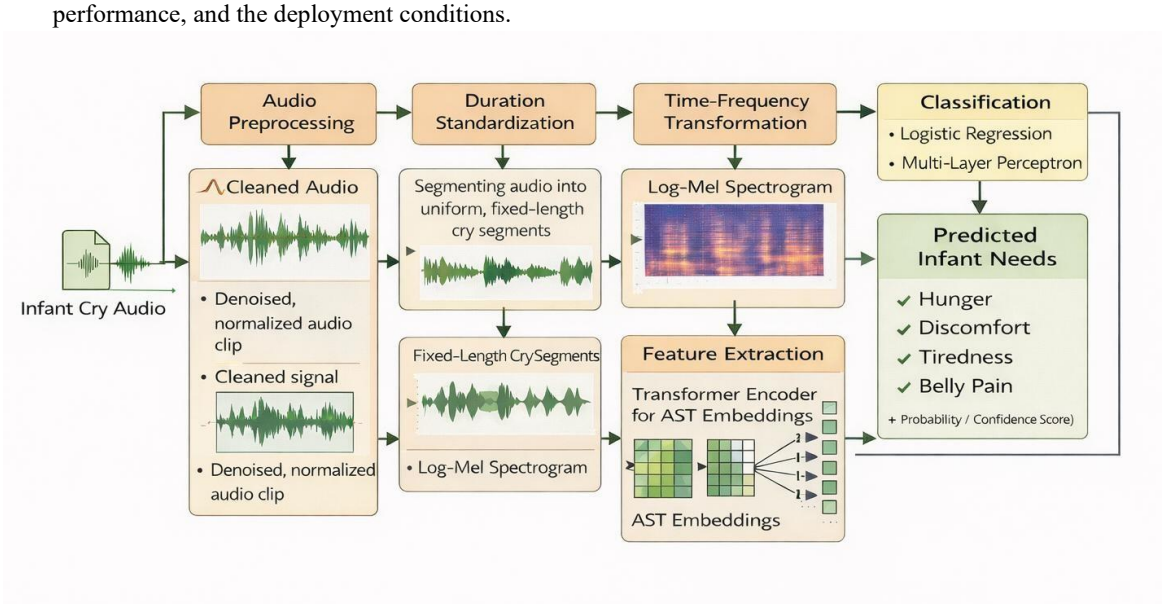


Fig. 2 Workflow diagram of proposed system

In the provided Table 1, we present crucial parameters for a Model Training and Classification along with recommended values to guide the optimization process.

Table 2 Classification Model Parameters and Suggested Values

Parameter	Range	Suggested Value
Audio Sampling Rate (Hz)	8,000 - 44,100	16,000

Cry Segment Duration (seconds)	1 - 5	2 - 3
Mel Frequency Bands	64 - 128	128
AST Embedding Dimension	256 - 768	768
Logistic Regression Regularization (C)	0.01 - 10	1.0
MLP Hidden Layers	1 - 3	2
MLP Hidden Neurons	64 - 512	128, 256
Learning Rate	0.0001 - 0.01	0.001
Batch Size	8 - 64	16, 32
Number of Training Epochs	20 - 100	50

In the audio sampling rate parameter, which is usually between 8000 Hz and 44 100 Hz, a standard of 16,000 Hz is suggested to trade off signal fidelity and computation with efficiency. In regards to the duration of cry segment, ranging between 1-5 seconds, it is recommended to use a constant duration of 2-3 seconds to have enough temporal detail and an equal length of input. The suggested number of Mel frequency bands, which can be either 64 or 128 is recommended to be 128 in order to obtain fine-grained spectral properties of infant cries.

On the AST embedding dimension (possibly 256 up to 768) it is suggested to use the full 768-dimensional representation to avoid losing rich acoustic information acquired during transformer pretraining. In the case of Logistic Regression regularization, the value of the parameter is recommended to start with 1.0, which will stabilize between bias and variance. The Multi-Layer Perceptron (MLP) case, it is recommended that we should use two hidden layers with 128 or 256 neurons to allow the effective non-linear modelling without too much computing complexity.

The learning rate of 0.0001 to 0.01 is suggested to start with 0.001 in order that the convergence is always steady throughout the training. Likewise, a batch of 16 or 32 is chosen out of the range of 8 to 64, which is an adequate compromise between training stability and memory efficiency. Lastly, the number of training epochs (between 20 and 100) is proposed to start at 50 epochs to enable adequate learning with the reduced chances of overfitting.

These are the important parameters that dictate the behaviour and performance of the infant cry classification system proposed. The suggested values provide a reasonable initial value and can be further changed based on the nature of data, how it performs at validation, and the purpose of its use.

The systematic framework of automated infant cry classification using transformer-based feature extraction and lightweight machine learning models are given in algorithm 1. Representation learning on ASTs is achieved by combining audio preprocessing, AST-based representation learning and efficient classification:

Algorithm 1: Infant Cry Classification for General Needs Using AST and Machine Learning Input:

-Infant cry audio dataset containing labelled cry samples for general needs (Hunger, Discomfort, Tiredness, Belly Pain)

- Audio sampling rate = 16 kHz $f_{s=16\,kHz}$
- Fixed cry segment duration = 2-3 seconds
- Pretrained Audio Spectrogram Transformer (AST)
- Classification models: Logistic Regression, Multi-Layer Perceptron (MLP) - Training–validation split = 80% training, 20% validation **Output:**
- Best-performing classification model for general infant need prediction - Predicted infant need category for each cry sample

Process Steps:

1. Initialization:

- Load the infant cry audio data.

$$\{(x_i, y_i)\}_{N\{i=1\}}$$

Where, x_i the input of the i-th sample. y_i the corresponding label of that sample (e.g., Hunger, Discomfort, etc.) i = 1 … N indicates the dataset contains N total samples

- Define preprocessing parameters including noise reduction, normalization, and resampling.
- Initialize the pretrained AST model for feature extraction.

- Initialize classification models: Logistic Regression and MLP with predefined hyperparameters.

2. Audio Preprocessing:

- For each infant cry audio sample $x(t)$:
- Remove silence segments.
- Apply noise reduction.
- Normalize amplitude levels.

$$x_{norm(t)} = \frac{x(t)}{\max(|x(t)|) +}$$

ε where, x(t) original audio signal at time t max(|x(t)|) maximum absolute amplitude in the signal ε (epsilon) small constant to avoid division by zero
x_norm(t) normalized signal

- Resample audio to a fixed sampling rate.
- Apply duration standardization to obtain fixed-length cry segments.

$$x_{\{std\}(t)} = x(t), ; \ t \in [0, T]$$

where, x(t) original signal
x_std(t) standardize signal
t ∈ [0, T] the signal is considered only within the time interval from 0 to T

3. Time–Frequency Transformation:

- Convert each standardized cry segment into a Log-Mel spectrogram representation.

$$S = \log(Mel(|STFT(x_{std})|^2) + \varepsilon)$$

where, x_std standardized audio signal
STFT(x_std) Short-Time Fourier Transform
Converts the signal from time domain to time-frequency domain
$|STFT(x_{std})|^2$ power spectrogram
Magnitude squared gives signal energy at each frequency
Mel(·) Mel filterbank
Converts frequencies to the Mel scale to match human hearing perception log(·)
logarithmic compression
Reduces dynamic range and highlights perceptually relevant differences, ε (epsilon) small constant Prevents taking log of zero (numerical stability)

- Ensure spectrogram dimensions are compatible with AST input requirements.

4. AST Feature Extraction:

- Pass the Log-Mel spectrograms through the pretrained AST model.
- Extract deep feature embeddings representing time–frequency cry characteristics.

$$z_i = AST(S_i), z_i \in \mathbb{R}^{768}$$

The feature embedding is defined as $\mathbf{z_i} = \mathbf{AST(S_i)}$, where $\mathbf{z_i} \in \mathbb{R}^{768}$ represents the 768-dimensional representation extracted from the Audio Spectrogram Transformer.

$$Z = [z_1, z_2, \dots, z_N] \text{ where,}$$

$z_1, z_2, \dots, z_n$ are individual feature vectors.
Each z_i is typically a fixed-length embedding (for example, the 768-dimensional vector you extracted using AST).
Z is formed by stacking all these vectors together.

- Store the extracted embeddings for classification.

5. Classification Model Training:

- For each classifier (Logistic Regression and MLP):
- Train the classifier using AST feature embeddings from the training dataset. $\hat{y} = f(Z; \theta)$

Where, The predicted label is computed as ŷ = f(Z; θ), where f denotes the trained classifier with parameters θ, operating on the feature matrix Z.

- Logistic Regression probability:

$$P(y = k \mid \mathbf{z}) = \frac{e^{\mathbf{w}_{kT}\mathbf{z}}}{\sum_j e^{\mathbf{w}_j^T \mathbf{z}}}$$

Where, z feature vector (embedding of the input audio) w_k
weight vector associated with class k
$w_k^T z$ dot product measuring how strongly the input matches class k
e^(·) exponential function to ensure positive scores
$\sum_j e^{(w_j^T z)}$ normalization term summing scores for all classes

P(y = k | z) probability that the input belongs to class k -

Validate the trained model using the validation dataset.

6. Evaluation:

- Evaluate model performance using classification metrics: Accuracy, Precision, Recall, F1-score - The F1-score, which provides a balanced measure of precision and recall, is calculated as:
- F1-score:

$$\text{Precision} = \frac{TP}{TP + FP}, \text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F1-score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Where:

TP: Correctly predicted infant need samples

FP: Incorrectly predicted infant need samples FN:

Missed predictions for a given infant need

7. Model Selection:

- Compare the performance of Logistic Regression and MLP models.
- Select the classifier achieving the highest F1-score as the final model.

$$f_{f} * = \arg\max(F1)$$

Where, f is a candidate model (classifier)

F1 is F1-score of that model (balances Precision and Recall) arg $\max_{(f)}$ the value of f that gives the maximum result f* the selected optimal model

8. Output:

- Use the selected model to predict general infant needs for unseen cry samples.
- Output predicted classes: Hunger, Discomfort, Tiredness, Belly Pain.

$$\hat{y} \in \{Hunger, Discomfort, Tiredness, Belly\ Pain\}$$

where, ŷ (y-hat) is the predicted label produced by the model ∈ "belongs to" or "is an element of"

{Hunger, Discomfort, Tiredness, Belly Pain} → the set of cry categories

4 Data Collection

The audio samples of infant cry utilized in this paper are the samples of infant cry recorded in the Donate-a-Cry Corpus, the publicly available dataset created by the manner of Donate-a-Cry mobile app on Android and iOS platforms. The corpus is made of infant cry recordings uploaded by users in their pure and unedited state and recorded in a real world. Audio files are accompanied by self-reported metadata contained in the file name that includes a unique identifier of the application instance, Unix time, version of the application, gender of the infants, age, and the perceived cause of the cry. The videos are delivered in CAF and 3GP, timestamps in iOS are stored in seconds and Android devices in milliseconds. There are categories of cry reason e.g. hunger, discomfort, pain, tiredness and fear, whereas age ranges fall in between newborns to toddlers. Considering that annotations are as well supplied by care givers, the dataset is subject to natural subjectivity and acoustic variations, which is ideal in developing powerful models. The entire data set may be denoted as D = (ai,mi) (i=1) N where aids the raw cry audio and m i is the related metadata. The data are published under Open Database License (ODbL 1.0), and the usage of the data in this paper does not violate any ethical or licensing restrictions.

Dataset link: https://github.com/Deekshita-G/General_Needs_Classification/blob/main/dataset

Table 3: Original Dataset: Class-wise Distribution of Cry Samples

Cry Class Categories	Count
discomfort	27
hungry	382
tired	24
burping	8

belly pain	16
Total Cry Samples	457

The value of number of audio samples per infant cry category in original dataset is given in table 2. Arrangement is skewed with regards to distribution. The busiest class is the hungry with 382 samples, meaning that the most common or annotated cries are associated with feeding. On the other hand, there are a lot less recordings in the category burping (8 samples), belly pain (16 samples), tired (24 samples), and discomfort (27 samples). Such unevenness may influence the process of training models since the machine learning classifiers will be biased towards the majority classes. In the absence of mitigation techniques (data augmentation, weighting the classes or resampling), the model can tend to be very high overall accuracy and extremely low on minor classes. Thus, the management of the imbalance of classes is a significant measure in enhancing the consistency of predictions among all needs of the infants.

4.1 Data Preprocessing

When dealing with every audio sample of infant crying, a series of preprocessing actions are performed in an attempt to enhance the quality of the signal and create consistency throughout the data set. Noise reduction is done after first removing silence segments and keeping only significant portions of the cry, which are then reduced to remove the background disturbance. The amplification of the audio signal is then adjusted to achieve a constant loudness in the audio signal throughout the various recordings. All samples are resampled then to a common sampling rate in order to normalize the time resolution. Lastly, standardization of the duration is used by either the trimming or padding of each signal to a constant duration to end up with uniform cry segments that are much more amenable to further feature extraction and classification steps.

Table 4: After Preprocessing and Feature Extraction both done in single code.

Metric	**Value**
Total usable samples (X, y)	457
Feature dimension	768
Unique classes	[2, 3, 4]

Table 3 displays, the dataset used for model training which is composed of 457 usable audio samples all of which are accompanied by their corresponding label (X, y). Each sample is then represented using a 768 - dimensional feature vector after preprocessing and feature extraction and this value matches the embedding size of the feature extraction model. There are three distinct classes in the label set, which were coded [2, 3, 4], meaning that not all original categories of cry were retained after a filtering/cleaning process. This can be because of the exclusion of underrepresented classes, the combination of categories or the elimination of noisy samples in order to enhance the reliability of the model. In general, these measures show that the training data is relatively large, high-dimensional in the representation, and specialized to a narrow set of infant-need categories, which can be used to enhance classification stability but at the cost of generalization to non-observed types of crying.

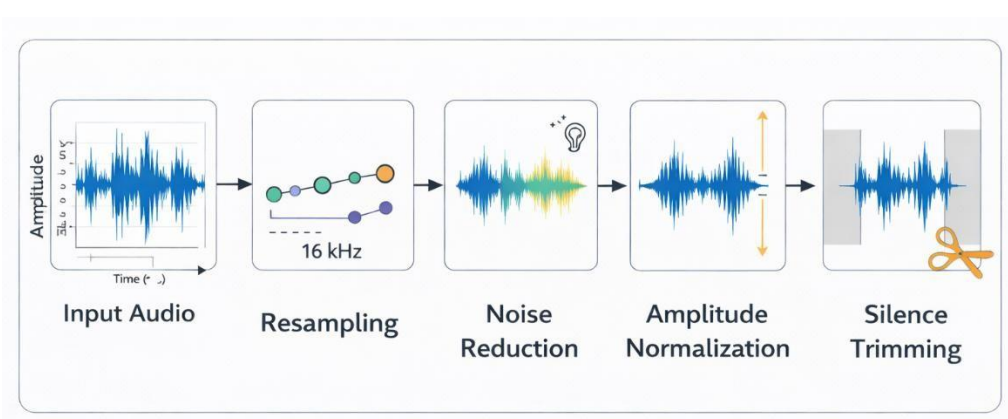


Fig 3: Audio Pre-processing flow

4.1a Audio Loading and Resampling

In fig 3, the first step in preprocessing is loading each infant cry audio file and converting it into a uniform sampling rate. Since recordings may originate from different devices, they can have varying sampling frequencies. To ensure consistency, all signals are resampled to a fixed rate (e.g., 16 kHz). A discrete-time audio signal can be represented as

$$x[n] = x(nT_s) \quad (1)$$

In equation 1, where $T_s = {}_{fs}$is the sampling interval and $f_s$is the sampling frequency. Resampling guarantees that every signal is represented with identical time resolution that is needed in the stable extraction of features and model training.

4.1b Amplitude Normalization

Audio recordings often vary in loudness due to differences in recording distance or microphone sensitivity. To reduce this variation, amplitude normalization is applied so that signals lie within a standard range. One normalization formula is the common normalization formula. A common normalization formula is

$$x_{norm}[n] = \frac{x[n]}{\max(|x[n]|)} \tag{2}$$

This operation in equation 2 scales the signal amplitude to a range of -1 to 1 without changing the spectral structure of the signal. Normalization enhances strength by ensuring that the model does not learn amplitude-related biases instead of learning acoustic patterns that have any meaning.

4.1c Silence Removal / Trimming

Silence may be in the recording of infant cry before or after the actual cry. Energy decay pruning allows the model to concentrate on the informative areas. Silence detection normally involves making use of an energy threshold:

$$E = \sum_{n=1}^{N} x[n]^2 \tag{3}$$

In equation 3Frames having energy less than a predetermined value are eliminated. The move is useful in decreasing redundant information, accelerating processing, and making sure that the features that are extracted are real cries and not silence of the surroundings.

4.1d Time–Frequency Transformation (Spectrogram Generation)

Here, the frequency spectrum of the signal is created by a spectrogram (also known as spectrogram generation). Time-frequency representations are much more effective as the input to machine learning models as compared to raw waveforms. As such, Short-Time Fourier Transform (STFT) is used to transform the signal into a spectrogram:

$$X(m,k) = \sum_{n=0}^{N-1} x[n]\, w[n-m]\, e^{-j2\pi kn/N} \tag{4}$$

The Equation 4, where $w[n]$ is the window function and $m$represents the time frame. The spectrogram shows how frequency content evolves over time and captures pitch, harmonics, and energy patterns that characterize infant cries.

4.1e Log-Mel Scaling

To approximate human auditory perception, the spectrogram is mapped to the Mel scale and log-compressed. The Mel frequency conversion is

$$Mel(f) = 2595\log_{10}\left(1 + \frac{f}{700}\right) \tag{5}$$

In equation 5, This transformation removes irrelevant frequencies to the perceptual system, and compresses the dynamic range. The reason why log-Mel spectrograms are popular is that they preserve significant acoustic patterns, yet become less sensitive to noise and amplitude change.

4.1f Feature Standardization

Normalization across the dataset is used by implementing standard scaling before the features are fed into the classifiers like the Logistic Regression and MLP:

$$z = \frac{x-\mu}{\sigma} \tag{6}$$

In equation 6 with new being the mean and sigma being the standard deviation. This guarantees that both the features and the mean and the unit variance are zero and stabilize the optimization, as well as the convergence, particularly in neural networks.

4.1g Transformer-Based Feature Extraction (AST Embeddings)

Lastly, the processed spectrogram is inputted to the Audio Spectrogram Transformer (AST) that generates a fixed-length embedding (768 dimensions) AST applies self-attention to capture long-range temporal and spectral dependencies:

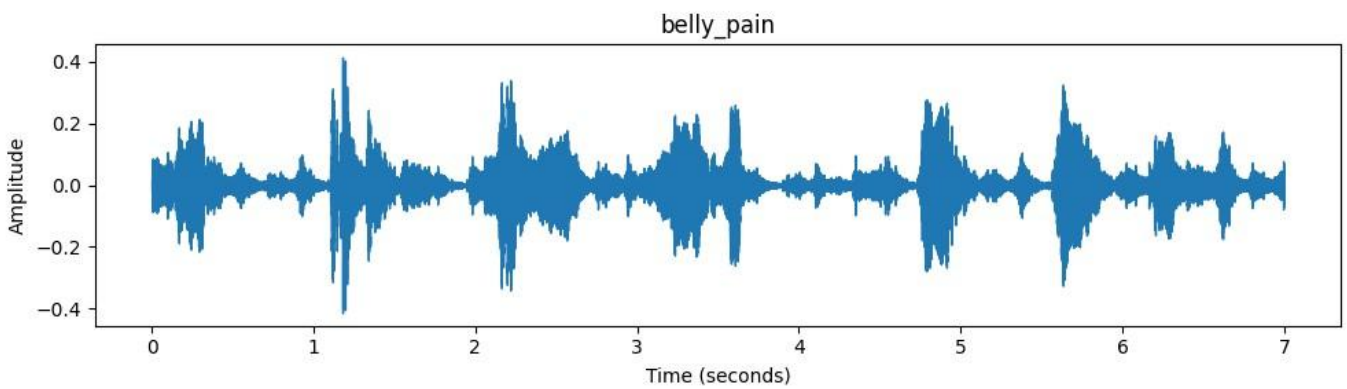
$$Attention(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (7)$$

The This mechanism in equation 7 enables the model to emphasize significant timefrequency areas of the cry signal to produce discriminative representation that is subsequently utilized in classification.

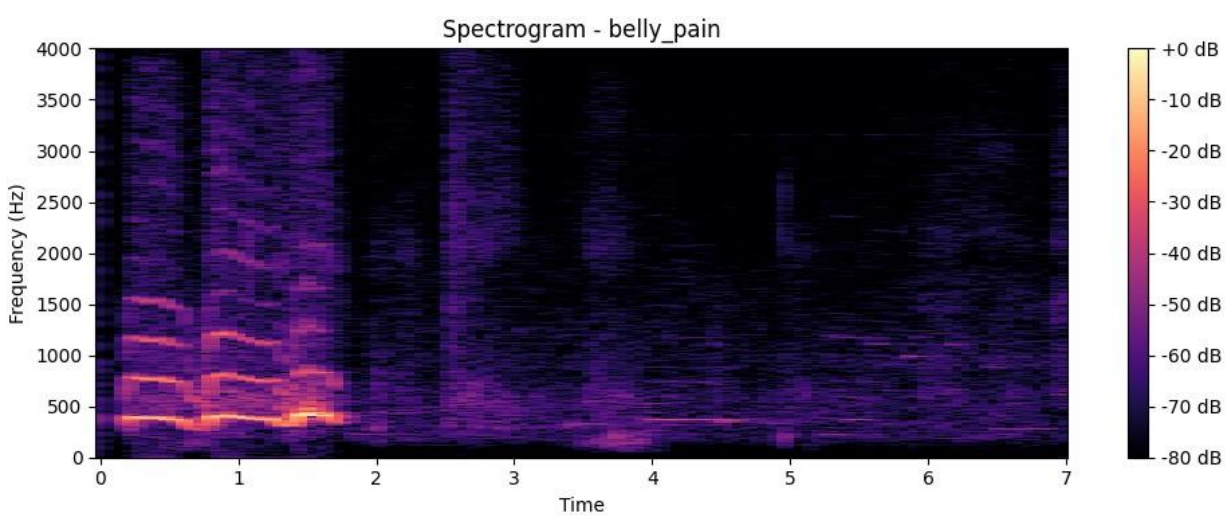
4.1h Visual examples of Cry Signal waveforms

This section reflects the visual display of infant cry signals both in time-domain waveforms and time-frequency spectrograms. The waveform plots show how the amplitude changed with time, and there were variations of hunger, discomfort, tiredness, belly pains and categories. These visual patterns are useful in recognizing specific acoustic pattern, including high repetitive peaks in hungry cries, disrupted patterns in discomfort cries, low-energy and spaced-out cries in tired cries, acute unstable peaks in belly pain cries and high starting intensity in cries.

The figure 4, is the time-domain signal of a baby cry that is termed as belly pain. Time is displayed in seconds on the horizontal axis and the amplitude on the vertical axis, which is the intensity of the audio signal. The waveform is irregular, high amplitude bursts with sharp peaks and different energy levels indicating the features of distress that are usually related to abdominal pain. The variation of amplitude and the intermittent lapses emphasize the imbalanced crying pattern, and it is appropriate to use it in extracting the acoustic features and further classification in the proposed infant cry detection system.



(a)

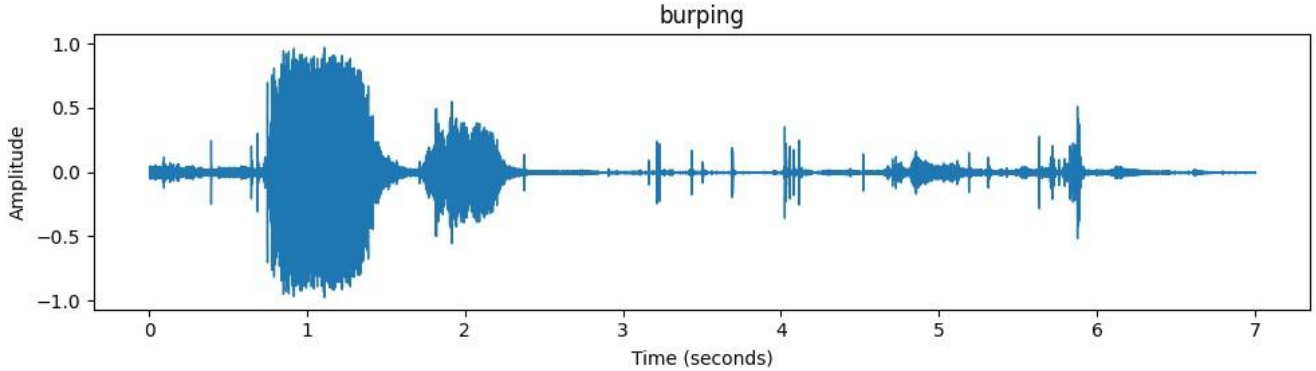


(b)

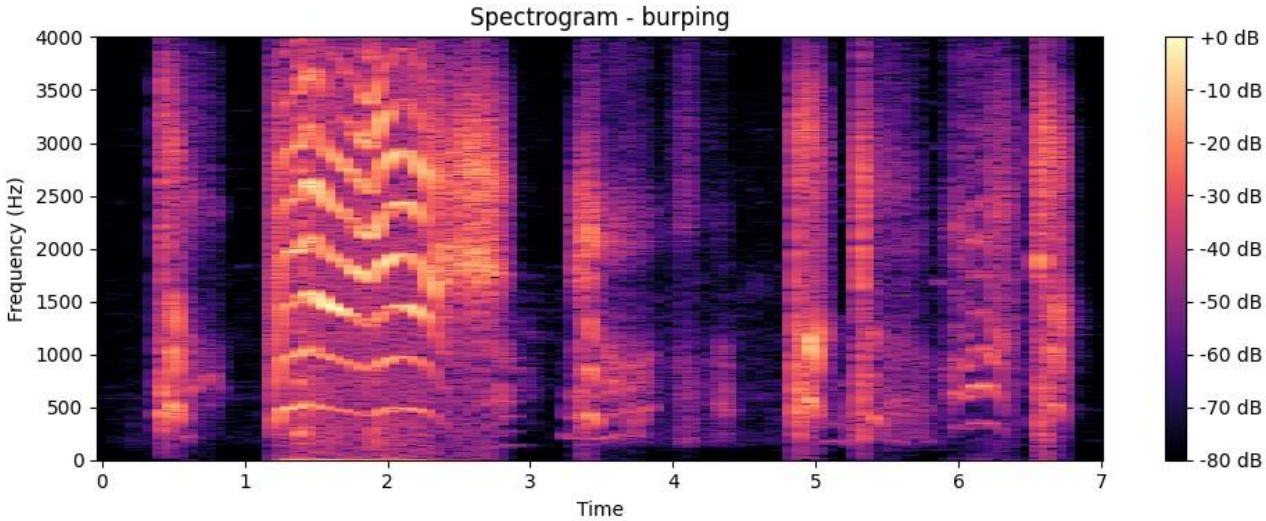
Fig 4: Belly pain infant cry: (a) the waveform representation, (b) the spectrogram.

The figure 5, is Burping waveform reveals a high amplitude burst at the beginning of the waveform which is superseded by a comparatively smaller amplitude signal activity with intermittent lapses with time. The amplitude is peak sharp within the first few

seconds, which is evidence of an abrupt release of air or discomfort, and then it crops out with some short spikes. In comparison with other types of cry, the energy distribution is more centred at the start of the cry with the subsequent long low-amplitude bouts, which shows a short but strong crying habit often linked to burger pain.



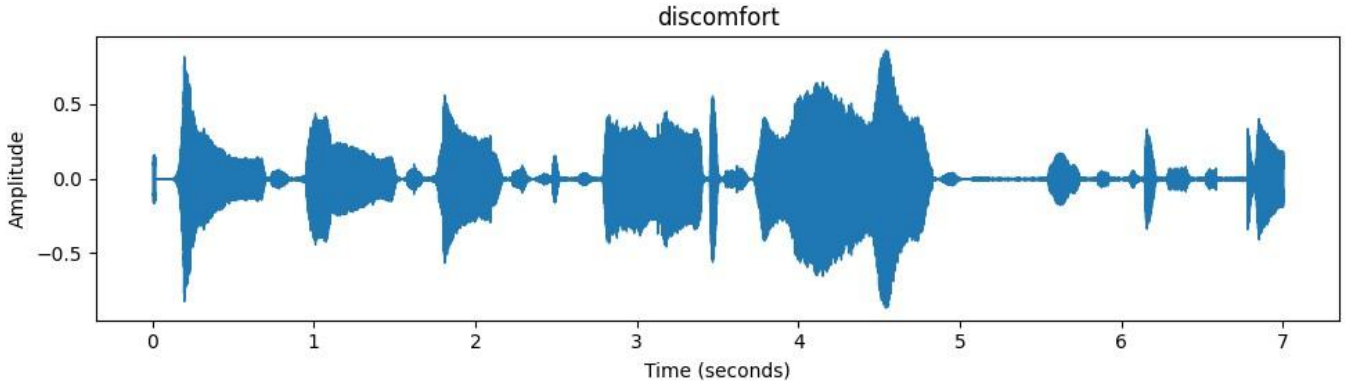
(a)



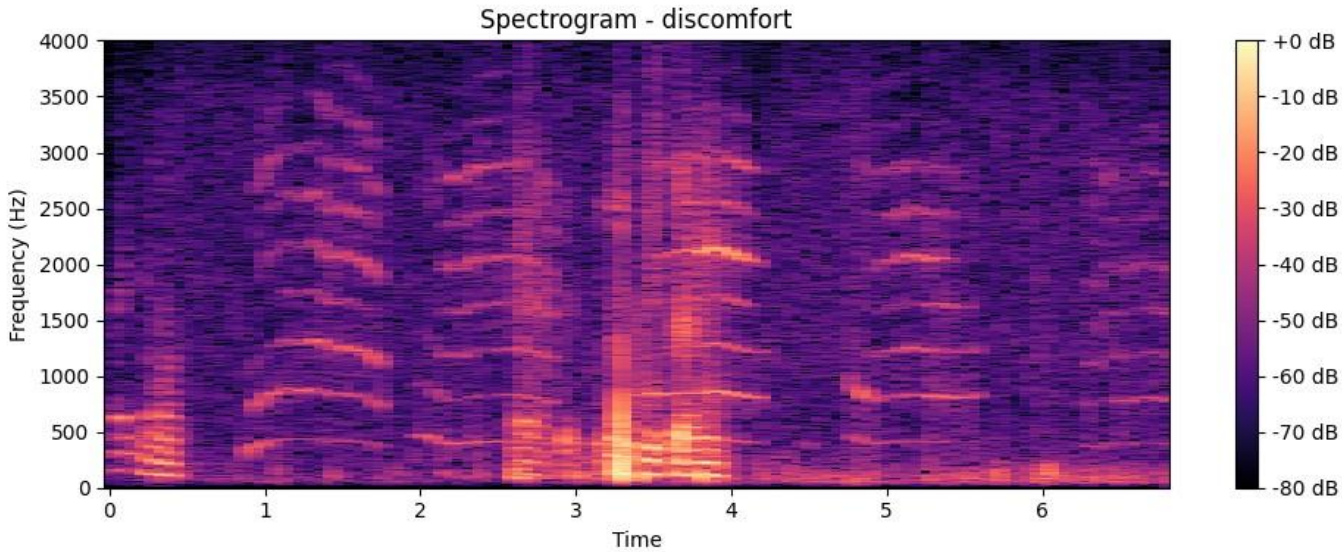
(b)

Fig 5: Burping infant cry: (a) the waveform representation, (b) the spectrogram.

Figure 6, is discomfort waveform displays medium to high amplitude changes that vary unequally in the duration of the waveform. There are repetitive bursts with pronounced variation and temporary halt in the signal, which shows that everything in it was restless. The amplitude peaks are common but not as severe as hungry cry, which indicates that there is a constant level of distress that is moderate and that corresponds to overall discomfort states in infants.



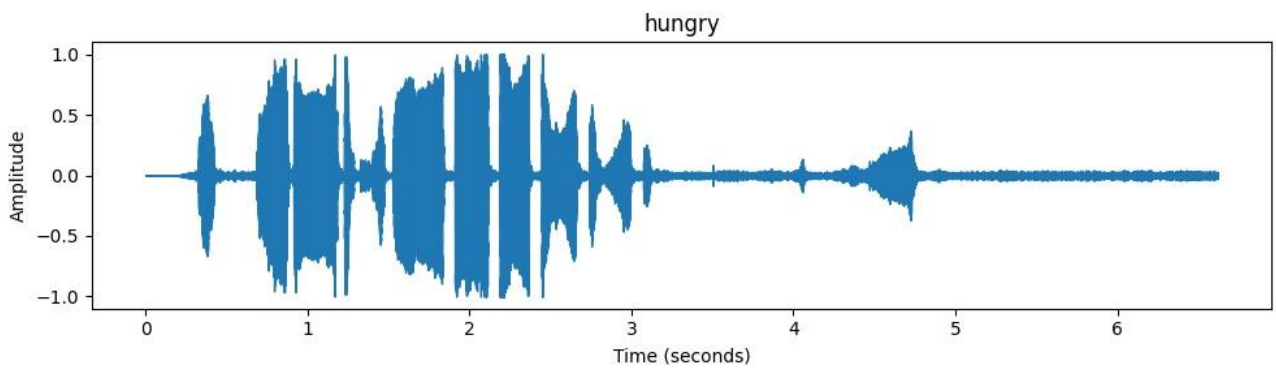
(a)



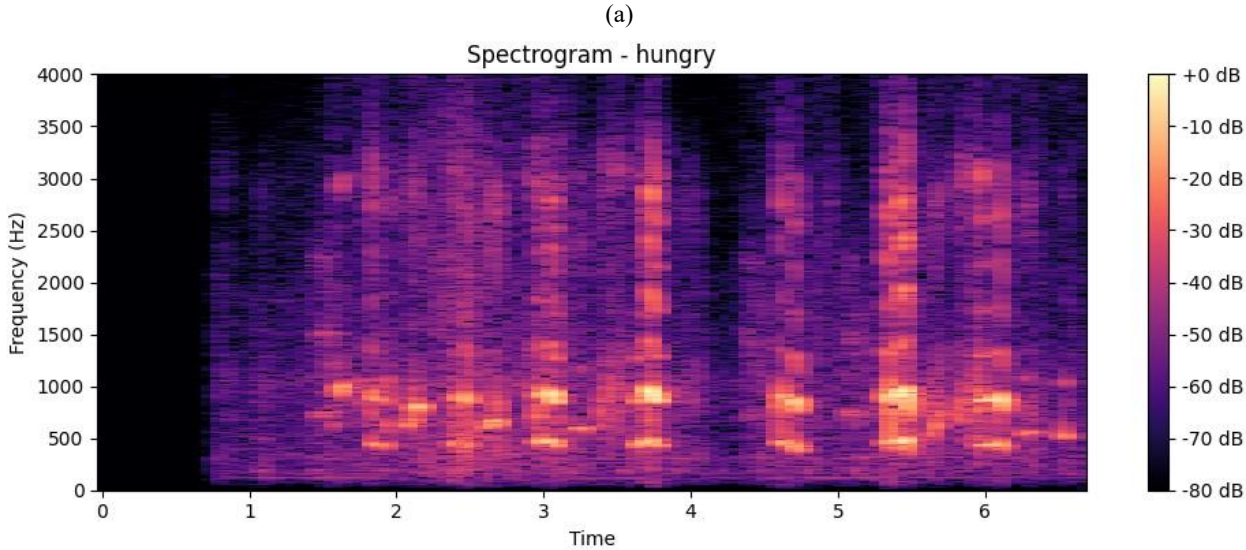
(b)

Fig 6: Discomfort infant cry: (a) the waveform representation, (b) the spectrogram.

The figure 7, is Hungry waveform is defined by intense, periodic, and intense bursts particularly in the beginning and middle of the signal. The amplitude is often of high peak value and lots of silent intervals, which is a sign that the crying is very urgent and rhythmic. It is a regular and pronounced pattern that is indicative of demand-based cry, which is normally a cry of hunger and thus it is distinguishable since it is sustained with energy and repetition.



(a)

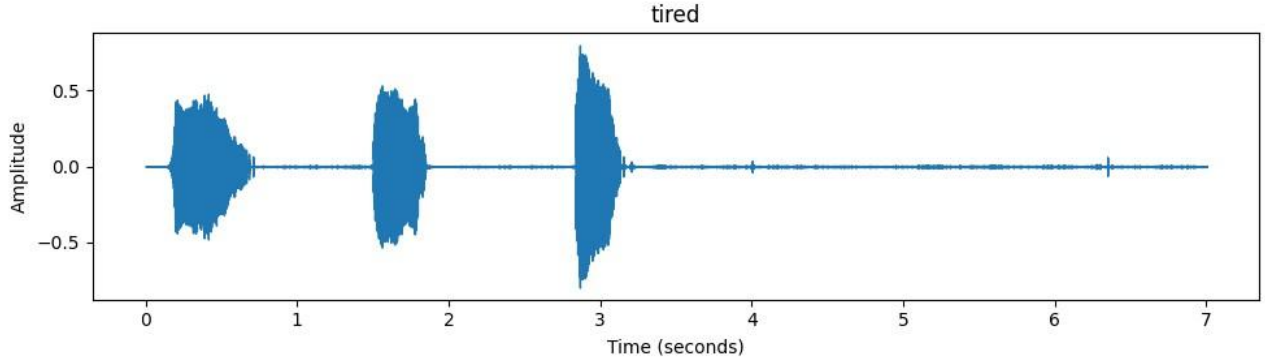


(b)

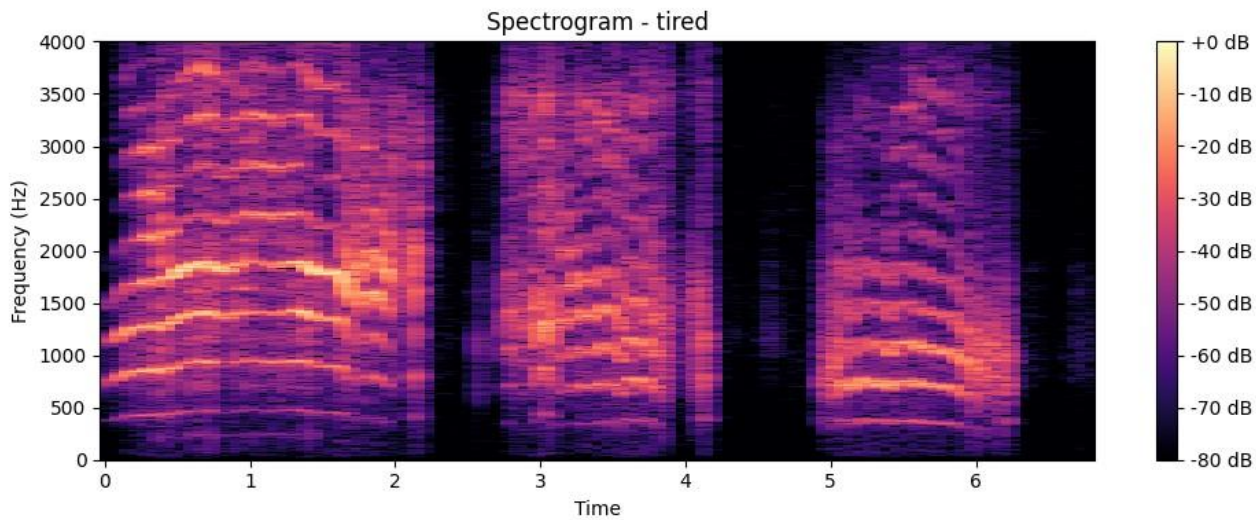
Fig 7: Hungry infant cry: (a) the waveform representation, (b) the spectrogram.

The figure 8, is tired waveform contains fewer bursts of high amplitude with increased periods of silence or low energy in between.

The segments of cry are less severe and intense than those of hungry or discomfort cries and have a less intense pattern, more distant. The diminished change in amplitude and the prolonged silence phases imply the decreased energy production, which is usually related to crying related to fatigue among infants.



(a)



(b)

Fig 8: Tired infant cry: (a) the waveform representation, (b) the spectrogram.

4.2 Data Deduplication and Unique Sample Identification

In order to guarantee the integrity and reliability of the learning process, there was a data deduplication step done before the training of the model. The datasets of infant cry may have duplicated or highly similar audio files as a result of overlapping pieces, uploads of the same piece multiple times, or silence-padded recordings. This redundancy may result in a model that is biased and unnaturally inflated in terms of performance. To cope with this, the entire audio samples were analyzed to detect and eliminate duplicate or similar audio samples of cry. Unique and valid cry samples were only retained and subjected to a subsequent processing. This process will guarantee that every feature vector is associated with a separate acoustic event and will enhance generalization of the model and avoid overfitting due to similarities between them. The original dataset can be defined as equation 1:

$$\mathcal{D} = \{(x_i, y_i)\}^{N}{}_{i=1} \tag{8}$$

where $x_i$represents an infant cry audio sample and $y_i$denotes its corresponding class label.

Due to repeated recordings or overlapping audio segments, duplicate or near-duplicate samples may exist. A deduplicated dataset $\mathcal{D}_{\text{unique}}$is constructed as below equation 2:

$$\mathcal{D}_{\text{unique}} = \{\ (x_i, y_i)\ ||\ x_i \neq x_j\ \forall i \neq j\ \} \tag{9}$$

In a unique manner only, unique cry samples are kept, which implies that there is one-to-one mapping of acoustic events to feature vectors. This process avoids biases in models due to multiple samples and improves the performance of generalization.

Table 5: Unique and valid cry samples

Dataset Part	Unique Labels
y	[2, 3, 4]

y_test	[2, 3, 4]

The table 4, shows the different names of classes present in the training labels (y) and the testing labels (y test). The labels of the two splits are [2, 3, 4] that is to say, the three classes of infant-cry are available in both training and testing sets. This kind of consistency ensures that the data division was performed correctly, and the model is tested on the analogous set of classes it has been trained on. This uniformity of training and testing labels is essential in attaining quality performance assessment in the sense that it prevents the unseen-class problems and ensures quality comparison between the accuracy of prediction.

4.2a Class Imbalance Handling

The initial data is defined with the high imbalance of classes as such that there were certain groups of cries with a very low sample number in contrast to others. The inequality in the classes can result in the unequal classifiers which will favor the dominating classes and disregard the minor groups. To minimize this, class-balancing technique was employed in the training of the models. The base Logistic Regression model was class-weight balancing in which the less representative classes were given larger weights in the loss equation. This approach would ensure that all of the classes are fairly involved in the process of optimization and improved recollection of the minority groups where C represent the set of class labels, c k, C = c1, c2, c3, c4, cK, and each class ck has n k samples in equation 3:

$$n_k = |\{x_i \mid y_i = c_k\} \tag{10}$$

Class imbalance occurs when in equation 4:

$$n_k \ll n_j \text{for some } k \neq j \tag{11}$$

To mitigate this, class-weighted learning is applied during model training. The weight assigned to class $c_k$is defined as equation 5:

$$w_k = \frac{N}{K \cdot n^k} \tag{12}$$

These weights are incorporated into the loss function, ensuring that minority classes contribute proportionally during optimization and reducing prediction bias toward majority classes.

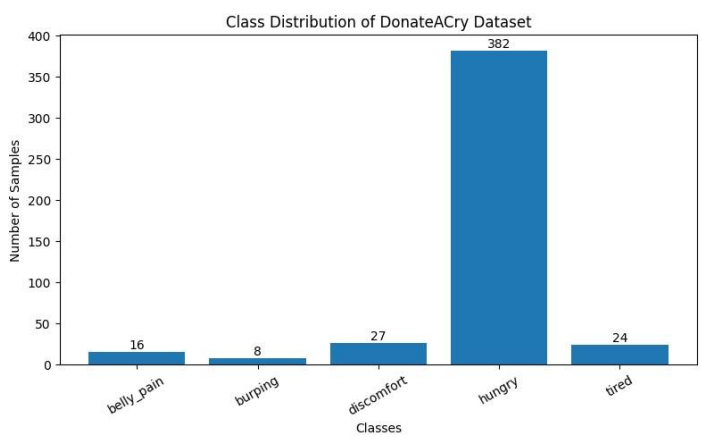


Fig 9: Class Distribution of dataset before balancing

The table 6, shows the calculated weights of the classes in which training of the model is done to overcome the issue of class imbalance in the data set. As certain categories of infant cry are much more common than others, when weights are applied to the minority classes, the classifier will give them more consideration when learning. The most dominant category, the hungry one, is assigned the lowest weight (0.2393), which contributes to the loss function less. On the contrary, we give relatively heavier weights to minority classes like burping (12.1667) and belly pain (5.6154) to make misclassifications in such classes punishable. Such classes as discomfort and tired are assigned moderate weights as they are intermediate in the dataset. With such weights the training process will be more balanced enhancing the ability of the model to predict the less frequent yet clinically significant need of infants as opposed to biasing towards the majority class.

Table 6: Class Balancing

Class ID	Class Label	Class Weight
0	belly_pain	5.6154
1	burping	12.1667

2	discomfort	3.3182
3	hungry	0.2393
4	tired	3.8421

4.2b Class Merging Based on Acoustic Similarity

Acoustic feature analysis revealed high similarity between the *belly_pain* and *burping* cry categories, combined with limited sample sizes. To improve statistical reliability, these classes are merged into a single category termed discomfort:

$$\{\text{belly pain,burping}\} \rightarrow \text{discomfort}$$

Formally, the new label mapping function $f(y)$is defined as:

$$f(y) = \begin{cases} \text{discomfort}, & \text{if } y \in \{belly\ \ pain, burping\} \\ y, & \text{otherwise} \end{cases} \quad (13)$$

The change also minimizes inter-class confusion by decreasing the number of classes in total to K=3 to maximize classification stability. The step removes differences that are present between recording equipment and formats and allows the downstream model to be focused on the important acoustic patterns rather than on sampling variations.

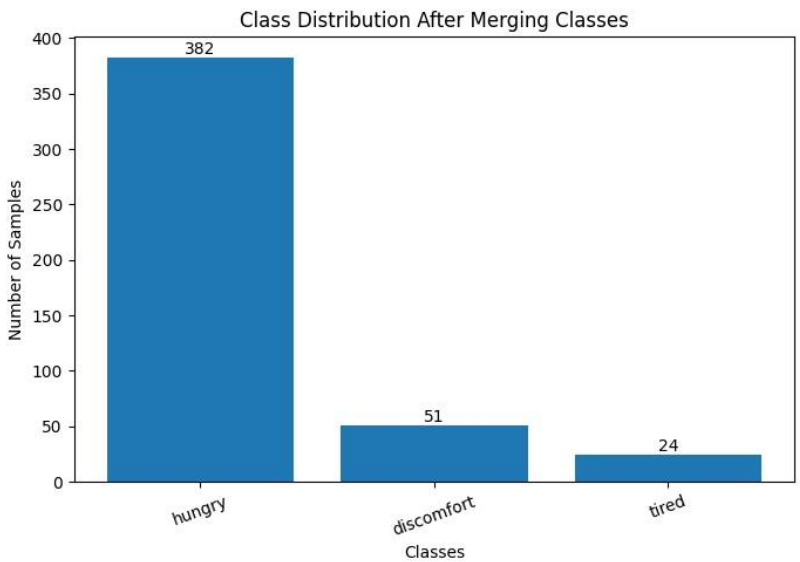


Fig 10: Class Distribution of dataset after balancing

4.3 AST-Based Feature Extraction

The Audio Spectrogram Transformer (AST) is a key factor in the infant cry classification system proposed since it can extract deep features of audio signals successfully. AST is a transformer-based architecture that is directly targeted to audio classification. In contrast to the conventional convolutional neural networks that pay much attention to local features pattern, AST employs a self-attention model to learn both short-term and long-term dependencies in the audio signal. It is especially appropriate to the study of complex acoustic events like baby cry, which commonly change with time and frequency, pre-processing and resampling of raw infant cry recordings is done to make them the same. The audio signals are converted to log – Mel spectrogram representations which are representations that are effective in maintaining both frequency and time-domain information. Such spectrograms are broken into fixed size patches and processed by the AST model. AST learns discriminative high-level embeddings through the application of several transformer encoder layers and multi-head self-attention, which are the underlying attributes of each category of cry. The results of the embeddings of AST are deep feature vectors used in the classification step. The system can record even the slightest changes in acoustics linking them to various needs of the infants like hunger, discomfort, fatigue, or pain by applying transformer-based feature extraction. This is good in terms of overall representation capability of the model and also classification performance. Audio Spectrogram Transformer (AST) is utilized in order to get deep embeddings of the log-mel spectrograms. AST implements a self-attention process that is used to model the long-term temporal dependencies of the cry signal and allow the model to take into account small-scale acoustic variations linked to various infant states.

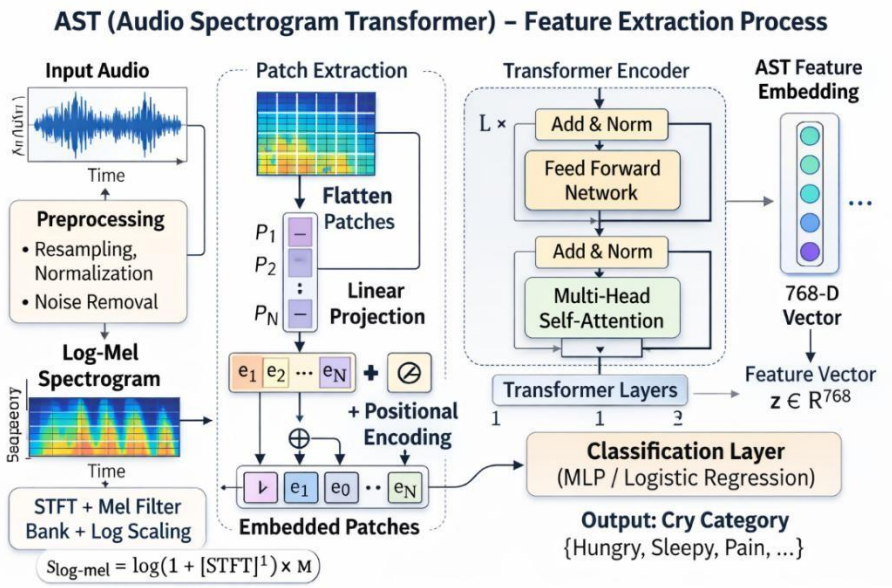


Fig 11: Working flow Audio Spectrogram Transfer for feature extraction process

4.3a Audio Input and Resampling

The raw records of infant cry are loaded and made standard to a standard 16 kHz sampling rate in the first stage. Resampling makes sure that all audio signals are time-resynchronized and this is critical to uniform feature extraction. This action removes the variations that are present between recording devices and formats, and the downstream model is focused on the important acoustic patterns rather than sampling variations.

4.3b Time–Frequency Transformation (Spectrogram Generation)

The division of the signal into time-frequency components and the lifetime of these components is known as time-frequency transformation (Spectrogram Generation). This is then accompanied by conversion of the audio waveform to a time-frequency representation, which is the Short-Time Fourier Transform (STFT). This converts the signal in the time domain to the spectrogram indicating the time-frequency variations of frequencies constituents. The STFT is computed as

$$X(t,f) = \sum_{n=0}^{N-1} x[n]\, w[n-t]\, e^{-j2\pi f n/N} \quad (14)$$

w[n]is the window function, frepresents frequency bins and where is the signal. This representation is significant as the features of infant cry are more accurately defined in the spectral representation than raw waveforms.

4.3c Log-Mel Spectrogram Conversion

The conversion of the spectra to log-Mel spectrograms was performed by the following formula: The spectrogram is then mapped to Mel scale which is approximating the aural sense of the human being by giving more weight to lower frequencies compared to the higher frequencies. Once the Mel filtering has been used to give the Mel frequency conversion, and the dynamic range is reduced and the finer spectral variations given emphasis with the help of logarithmic scaling. A perceptually meaningful representation of the logMel spectrogram that reflects the presence of pitch, tone, and energy changes in infant cries is achieved as a result.

$$M(f) = 2595\log_{10}\left(1 + \frac{f}{700}\right) \quad (15)$$

4.3d Patch Embedding for Transformer Input

The log-Mel spectrogram is split into small patches in the same manner as vision transformers. The patches are flattened and projected into a fixed-dimensional vector space. This puts the 2D spectrogram in the form of a sequence of embeddings, allowing the transformer model to act on audio as a sequence of tokens. Positional encodings are introduced to allow the model to maintain the information of time sequence.

4.3e Transformer Encoding (AST Feature Learning)

The patch embedding sequence is fed through many layers of transformer encoders comprising of self-attention networks and feedforward networks. Self-attention calculates interrelationships among all patches at the same time, which enables the model to accommodate long-range temporal and spectral relationship interdependence. The attention process can be characterized as

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (16)$$

where Q, K, and Vrepresent query, key, and value matrices. This stage enables the model to learn rich acoustic representations that encode complex cry patterns.

4.3.f. Global Embedding Feature (Aggregation of features)

Lastly, there is a summation of the transformer outputs into one fixed length embedding (768 dimensions in this work). This embedding is the summary of the complete cry signal into a small representation that contains both the information about time, spectrum and the context. These embeddings serve as high-level acoustic features and are used as input for downstream classifiers such as Logistic Regression and MLP for infant need prediction. Considering an input audio signal x(t) the log-mel spectrogram given by equation 7:

$$S = \log\left(\text{MelSpectrogram}(x(t))\right)$$

(17)

Where:

$x(t)$ represents the input audio waveform of the infant cry in the time domain, where $t$denotes time. Mel Spectrogram$x(t)$ denotes the Mel-scale spectrogram computed from the input audio signal $x(t)$, which represents the distribution of signal energy across Melfrequency bands over time. Log () Indicates the logarithmic compression applied to the Mel spectrogram to reduce dynamic range and emphasize perceptually relevant acoustic features. S denotes the log-mel spectrogram derived from the input audio signal.

$$S \in \mathbb{R}^{T \times F} \quad (18)$$

where:

$T$is the number of temporal frames and $F$is the number of Mel frequency bins. Patches Refers to non-overlapping or overlapping 2D segments extracted from the log-mel spectrogram S, which serve as input tokens to the transformer model. Transformer encoder layers A stack of self-attention–based layers used to model long-range temporal dependencies within the cry signal.

The spectrogram is segmented into patches and passed through multiple transformer encoder layers, producing a sequence of contextual embeddings:

$$H = \{h_1, h_2, \dots, h_T\}$$

(19)

Where:

Represents the sequence of contextual embeddings produced by the AST after processing the spectrogram patches. h_t Denotes the context-aware feature embedding corresponding to the t^{th}temporal frame, capturing both local acoustic properties and global temporal context.

T Denotes the number of temporal frames (or tokens) in the spectrogram after patch segmentation and transformer processing.

Table 7: Audio samples are into x and y labels.

Metric	Value
X shape	(457, 768)
y max label	4
Skipped files	0

The table 7, represents feature matrix X, is of shape (457, 768), which implies that there are 457 audio samples in the dataset, with each one having a 768-dimensional feature vector at the feature generation step of the process. This establishes that any recording that could be used was effectively converted to homogeneous numerical format that could be used to train a model. The largest value of label in y is 4 and this is as expected by the label encoding scheme applied in the dataset, as the class IDs reach to a maximum of 4. It aids in ensuring that the indexing of labels was not lost during preprocessing and filtering. Moreover, feature extraction did not exclude any files, and this means that the preprocessing pipeline has managed to process all audio samples without corruption, format errors, and extraction errors. This indicates high quality of data and a consistent process of features extraction.

4.4 Temporal Mean Pooling

Since infant cry recordings vary in duration, a fixed-length representation is required for downstream classification. Temporal mean pooling is applied across the AST output embeddings to aggregate contextual information into a single vector

$$\bar{e} = \frac{1}{T}\sum_{t=1}^{T} h_t \quad (20)$$

Where:

e Denotes the final pooled embedding vector, which captures the global acoustic characteristics of the infant cry and is used for downstream classification. $\frac{1}{T}$ Serves as a normalization factor which averages the embeddings such that a fixed length representation is obtained no matter the length of the input.

$\sum_{t=1}^{T} h_t$ Represents the total of frame-based embeddings throughout the commencing length of the audio signal.

T Refers to the sum of the number of temporal frames (or embeddings) of the AST output of a particular cry recording .

h_t The contextual embedding of the AST model of the temporal frame of infant cry signal.

The pooling strategy is a strong summary representation of the cry signal whilst being less sensitive to sporadic noise and periods of silence. The resultant embedding still contains global acoustic properties of infant cry classification.

4.5 Feature Matrix and Label Vector Construction

All pooled embeddings are concatenated to form the feature matrix:

$$X \in \mathbb{R}^{N \times 768} \tag{21}$$

Where:

X Represents the feature matrix formed by concatenating the pooled embeddings of all audio samples. e Denotes the fixed-length pooled embedding vector obtained for each infant cry recording after temporal mean pooling.

$X \in \mathbb{R}^{N \times 768}$ Indicates that the feature matrix has Nrows, corresponding to the total number of valid audio samples 768 columns, corresponding to the embedding dimension of the AST model

N Denotes the number of infant cry audio samples used for model training and evaluation. 768 Represents the dimensionality of the AST output embedding, as defined by the transformer architecture.

The corresponding class labels are stored in a target vector:

$$y \in \mathbb{R}^{N} \tag{22}$$

Where:

y Denotes the target label vector containing the ground-truth class labels for each audio sample.

$\mathbb{R}$ denotes the set of real numbers. It means that all elements in the matrix or vector are real-valued (not complex or categorical). It simply specifies the data type/domain of the values used in your feature matrix and label vector.

$y \in \mathbb{R}^{N}$ Indicates that the label vector contains one label per audio sample, aligned with the rows of the feature matrix X. Strict shape validation is performed to ensure dimensional consistency across all feature vectors prior to model training.

Table 8: Label Vectorization

Index	Label Name
0	belly pain
1	burping
2	discomfort
3	hungry
4	tired

The table 8, codes the numerical values of the various classes to the infant-cry groupings that had been used in the dataset. Each of the classes will be identified by a distinct integer number ranging between 0 and 4 that would represent five different needs of an infant belly pain, burger, sore, hungry, and sleepy. Under this indexing scheme, the machine learning model can accept categorical labels in numbers in training. It also renders things identical regarding the weight of the classes, calculations of the predictions and the model output can be decoded to meaningful classes names that can be interpreted in the final system. In order to not to label errors and ensure that the indexes predicted are correct according to the projected infant-cry categories in testing and deployment, it is important to maintain this systematic mapping.

4.6 Train-Test Data Splitting

The entire data set is split into training and testing sets with the 80:20 ratio:

$$D = D_{\text{train}} \cup D_{\text{test}} \tag{23}$$

where:

D Represents the entire data set with all sets of feature vectors and the class labels associated with them. D train Represents the training part of the data set, the one that is used to learn the model parameters.

Symbolizes the testing part of the data set, it is applied to test the performance of the trained model on unknown data.

∪ Denotes union operation, which implies that the entire dataset Dis consisting of the training and testing subsets.
$D = D_{\text{train}} \cup D_{\text{test}}$ Represents that the set of data is split into two distinct subsets that do not overlap each other and are used to comprise the entire set of data. 80:20 split Stipulates that 80 percent of the samples should be assigned to D train and 20 percent to D test. The stratified sampling is used to maintain class balance in both subsets that guarantee an impartial assessment and good generalization capacity.

Table 9: Labels after training and testing.

Dataset	Number of Samples
Training samples	365
Testing samples	92

The dataset in the table 9 was further broken into training and testing subsets to objectively assess the model performance. Among the entire samples, 365 recordings were utilized to train the model, which became able to learn about patterns, which were related to various categories of infant cry. The rest 92 recordings were used to test, which will offer an invisible dataset to measure the ability of the trained model to generalize to new data. This division makes sure that evaluation metrics are based on the actual predictive power of the model as opposed to one with a memory of the training data. To keep model robustness and avoid overfitting, it is crucial to have a test set that is dedicated to the validation of model robustness.

4.7 Logistic Regression Baseline Classification

The Logistic Regression is a supervised machine learning that is commonly used in classification. It approximates the likelihood of a given input to fall within a specific group with logistic (sigmoid) model. In contrast to linear regression, a continuous value, logistic regression predicts probabilities of the classes and labels them according to a decision threshold. In the case of multi-class problem it is normally extended either by the One-vs-Rest (OvR) or multinomial (SoftMax) strategy. One of the baseline classifiers applied in the infant cry classification was Logistic Regression. Following the derivation of high-level deep embeddings of the Audio Spectrogram Transformer (AST), the resultant feature vectors were scaled to the norm to make the feature vectors have the mean zero and the unit variance. These features, which were normalized, were then fed into the Logistic Regression model. As the work is concerned with a variety of types of cries (e.g., hunger, discomfort, tiredness, and so on), the multi-class logistic regression model was used. The model is trained to learn a set of weights, which is used to estimate the probability of classes as a linear combination of the AST feature embeddings. It minimizes the cross- entropy loss function to reduce the error in classification during the training process. Although being a relatively simple yet computationally efficient model, the high-quality feature representations produced by AST made Logistic Regression a successful one. This shows that robust feature extraction can be a great boost to even conventional machine learning classifiers.

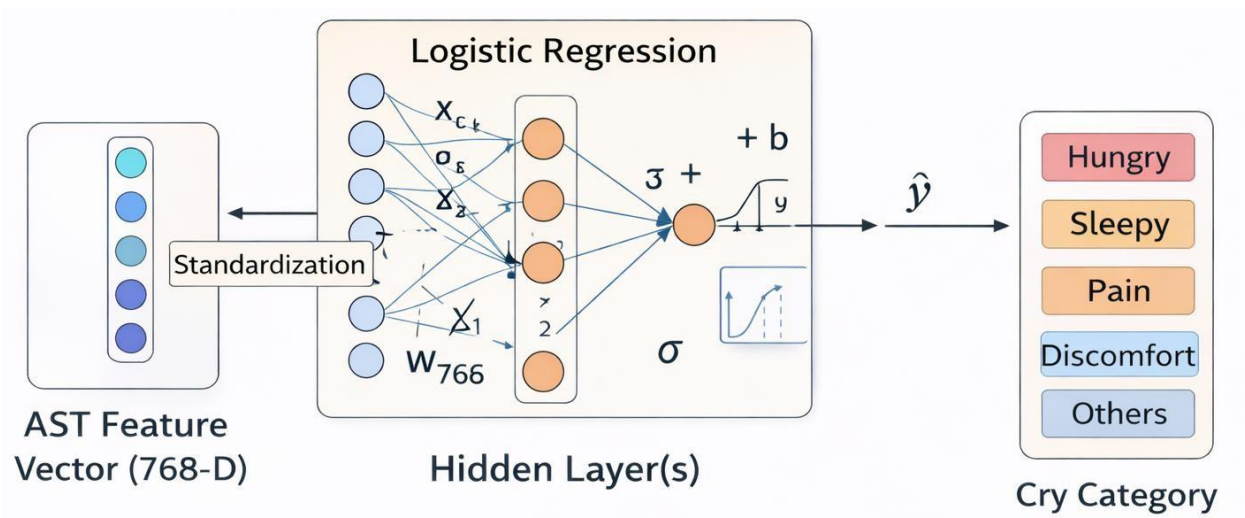


Fig 12: Logistic Regression Architecture

Table 10: Architecture of Logistic Regression

Layer	Type	Units / Parameters	Activation	Regularization
1	Input Layer	768 features (AST embedding)	-	Feature scaling applied

2	Linear Output Layer (Dense)	Number of classes (k)	Softmax	L2 regularization (optional)

The single linear transformation used in Logistic Regression is an application where the 768-dimensional AST feature set is transformed to class scores. Class probabilities are calculated by the softmax function with these scores and the most probable class is picked as prediction. The Logistic Regression does not have any hidden layers because it is a linear model. Regularization, e.g, L2 penalty can be used to avoid overfitting and enhance generalization.

4.7.a Softmax Function in Logistic Regression

This study will use Logistic Regression as a multi-class classifier over the AST feature embeddings. Given that infant cry recognition depends on more than two classes (e.g., hunger, discomfort, pain, tiredness), the model includes the extension of binary logistic regression into a multi-class context with the help of the softmax function. Given an input feature vector z, the model is used to compute a linear score in each of the classes based on learned weights and biases. The results are then fed into the Softmax function so as to transform the scores into normalized probabilities of the classes:

$$p(y = k \mid z) = \frac{ew^T kz+bk}{\sum_{j=1}^{K} ew^T jz+bj} \tag{24}$$

where $w_k$and b_k the parameters of the different classes are k, and K is the number of classes. The model model anticipates the need of the infant by choosing the is the parameter of class, is the parameter of class and is the number of classes. The model estimates the need of the infant by choosing the one with the largest softmax probability. It is a probabilistic formulation that allows interpretable predictions and also trains efficiently with low computational complexity, which makes Logistic Regression a powerful control group on the assessability of the separability of AST features.

```
LogisticRegression
LogisticRegression(class_weight='balanced', max_iter=1000,
                   multi_class='multinomial')
```

Fig 13: Logistic classification

4.8 Class Merging and Dataset Reconstruction

The primary performance evaluation showed that the minority classes like belly pain were affected by inadequate training samples. These classes are combined into one category of discomfort because of their similar sounds. The reorganization of the classes makes the total number of five classes become 3 and increases the statistical reliability of the situation and retains the meaningful differences in physiology. High similarity between the belly pain and burping cry categories was found with the acoustic feature analysis with limited samples. To enhance statistical validity, these classes are combined in one category named discomfort:

{belly pain,burping} → discomfort

Mathematically, the new label mapping f(y) is represented as follows as equation 25: discomfort,

$$f(y) = \begin{cases} \text{discomfort}, & \text{if } y \in \{belly\ pain, burping\} \\ y, & \text{otherwise} \end{cases} \tag{25}$$

The overall result of this transformation is the reduction of the total number of classes K=5 to K=3, which reduces the inter-class confusion and improves the stability of the classification.

Table 11: Labels after merging

Label ID	Class Name
2	discomfort
3	hungry
4	tired

The table 11, represents the final classes that were used in the model training and evaluation. The dataset has three categories of infantcry discomfort hungry and tired. The machine learning model works with these numerical IDs internally and the final system interprets a prediction using these class names.

Table 12: Finding Unique samples after merging

Variable	**Unique Values**
y unique	[2, 3, 4]
merged_y unique	[2, 3, 4]

The table 12, illustrates the distinct class labels in the original label vector (y) and the processed label set (merged_y). The distinct values in both instances are [2, 3, 4], which means that the three categories of infant-cry were maintained during preprocessing and label merging. This consistency will prove that there were no unwanted changes in labels during the preparation of the dataset. Maintaining identical label sets before and after processing ensures that the training targets remain valid and that the model learns from the correct class representations. After deduplication, class balancing, and class merging, the refined dataset is reconstructed as equation 26:

$$\mathcal{D}_{\text{final}} = \{(x_i, f(y_i))\}^{N}_{i='1} \tag{26}$$

Where:

D_{final} Denotes the final refined dataset obtained after deduplication, class balancing, and class merging.

$(x_i, f(y_i))$ Represents a feature–label pair for the i^{th}sample in the refined dataset.

x_i Denotes the feature vector corresponding to the i^{th}infant cry sample.

y_i Represents the original class label of the i^{th}sample before refinement.

$f(\cdot)$ Denotes a label transformation function that maps the original class labels $y_i$to the refined or merged class labels. $f(y_i)$ Represents the refined class label associated with the i^{th}feature vector after class merging or relabeling.

$i = 1,2, \dots , N'$ Indexes the samples in the refined dataset. N' Denotes the total number of valid and unique samples remaining after dataset refinement.

$N' \leq N$ Indicates that the number of refined samples is less than or equal to the original dataset size due to deduplication and filtering. where $N' \leq N$represents the number of valid, unique samples after refinement. By using this reconstructed dataset, feature normalization, training of the classifier and evaluation are made based on consistency and fairness across cry categories.

4.9 Multi-Layer Perceptron (MLP)

The AST model, in turn, in this project will encode every sample of infant cry audio into a 768-dimensional embedding vector, which captures its time-frequency properties. These embeddings were scaled using feature scaling so that they would have stable gradient updates during training. The normalized feature vectors were then inputted to a Multi-Layer Perceptron that comprised of one or more hidden layers with the ReLU activation functions enabling the network to acquire non-linear patterns in the cry representations. The last output layer provides the probabilities of every need category of an infant, using the Softmax function. The accuracy scores and classification reports were used to evaluate model performance as it was observed that the MLP successfully uses transformer-based embeddings to enhance the reliability of prediction compared to the baseline Logistic Regression model.

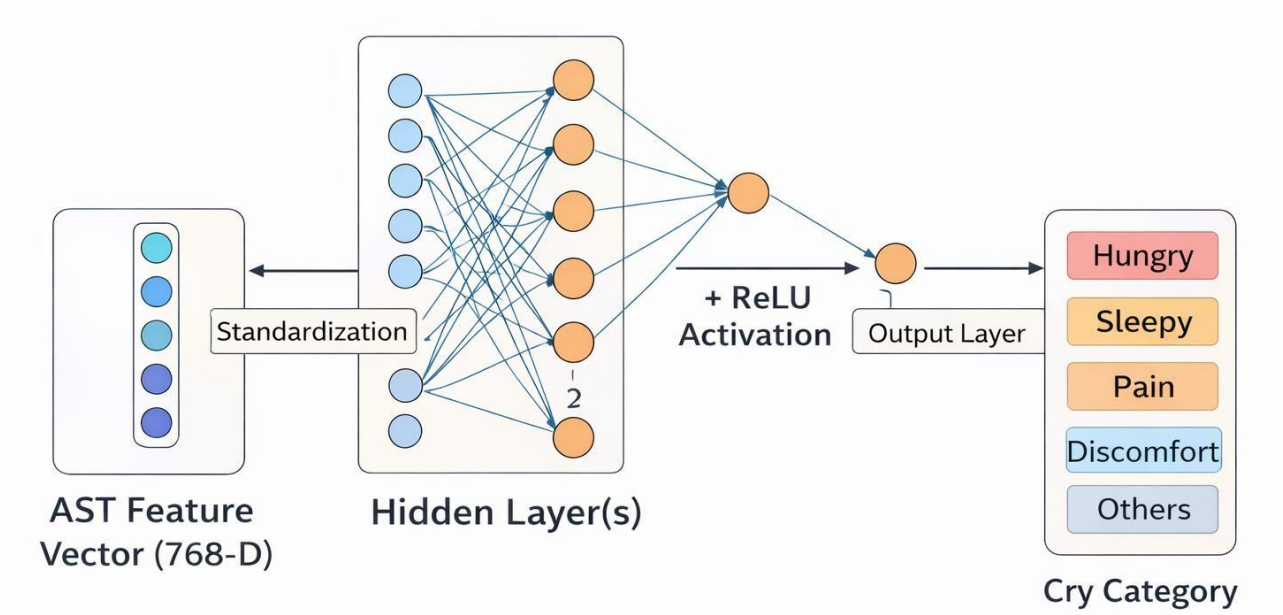


Fig 14: Working Design of MLP Classification

4.9a ReLU Activation Function

In the proposed MLP classifier, the Rectified Linear Unit (ReLU) activation function is applied in the hidden layers to introduce nonlinearity into the model. ReLU transforms the input signal by retaining only positive values while suppressing negative responses, which allows the network to learn complex relationships between AST feature embeddings and infant cry categories. It is mathematically defined as

$$\mathrm{ReLU}(x) = \max(0, x) \qquad (27)$$

where $x$is the linear combination of inputs and weights. The use of ReLU improves computational efficiency, accelerates training convergence, and helps prevent the vanishing-gradient problem. It is especially applicable to deep neural networks like the MLP that was employed in this study.

4.9b Softmax Function

Softmax is a function that is used in the last layer of the MLP classifier to transform the overall outputs of the network to the probability distributions of each infant-need category. This makes sure that the total of all the predicted values is one and that all the values are in the range of 0 to 1 and can be interpreted as probabilities. The Softmax function is defined as

$$(y = k \mid z) = \frac{ezk}{\sum_{j=1}^{K} ezj} \quad P \qquad (28)$$

where $z_k$represents the output score for class k, and Kdenotes the number of classes. The class with the highest Softmax probability is selected as the final prediction, making the function well suited for multi-class infant cry classification tasks.

Table 13: MLP Classifier Architecture

Layer	Type	Units / Parameters	Activation	Regularization
1	Input Layer	768 features (AST embedding)	-	Standard Scaling applied
2	Hidden Layer 1 (Dense)	256 neurons	ReLU	Dropout / L2 (if used)
3	Hidden Layer 2 (Dense)	128 neurons	ReLU	Dropout / L2 (if used)
4	Output Layer (Dense)	Number of classes (k)	Softmax	-

A Multi-Layer Perceptron (MLP) with two hidden layers consisting of 256 and 128 neurons is trained to capture nonlinear patterns in the AST feature space. ReLU activation functions are employed to enable efficient gradient propagation, and a softmax output layer is used for multi-class prediction.

```
MLPClassifier
MLPClasstfier(hidden_layer_sizes=(256, 128), max_iter=500, random_state)
```

Fig 15: MLP Classifier

4.9c Classifier Selection and Justification

The baseline classifier initially used in this study was Logistic Regression to determine the effectiveness of the AST embeddings to differentiate the infant cry categories in a linear feature space. Since the Logistic Regression learns linear decision boundaries, it is computationally inexpensive, understandable, and appropriate in determining whether the information derived through transformers already has discriminative information. Signals of infant cry are however very complicated and sensitive to minor spectral and temporal changes, and regularly lead to overlapping distributions of classes that cannot be completely detached using a linear model. MultiLayer Perceptron (MLP) classifier was proposed as a more expressive form in order to handle this limitation. The MLP can learn nonlinear decision boundaries as a result of hidden layers and activation functions and, therefore, is able to learn more intricate relationships in the AST feature space, unlike the Logistic Regression. It enables the model to distinguish better between the acoustically similar types of cries, which enhance classification performance without sacrificing the moderate cost of computation relative to more complex neural networks. Therefore, MLP is a logical evolution of the Logistic Regression, shifting towards non-linear modeling and retaining the portability and usability of the system.

5. Experimental Setup and Evaluation

The experiments were done to determine the efficacy of transformer-based acoustic representations with machine learning classifiers to categorize infant cry. The input was provided in the form of infant cry audio files of the Donate-a-Cry dataset. All audio samples were initially processed and turned into representations of spectrograms, and transformer-based feature extraction was performed to get embedding vectors of 768 dimensions representing fixed-length representations. The invalid or corrupt samples were eliminated in order to be consistent in feature dimensionality. The feature vectors obtained were split into training and testing groups to guarantee the fair evaluation. They used two classifiers, namely Logistic Regression and a MLP. In MLP model, the standard scaling was applied to all the features, and subsequently, training was done, which made the convergence stable. The comparison of the performance of the two models was done on a fair basis as both the models were trained on the same feature inputs. The predictions on the test set were estimated using the conventional classification measurements.in Metrics.

5.1 Partitioning Strategy of the Dataset

The processed dataset was split into training and testing parts to guarantee a solid evaluation. The division was done at the feature level following extraction of AST embedding in such a way that every sample in the audio had a single fixed-length vector. To maintain the distribution of infant cry classes in the two subsets a stratified partitioning method was employed. This will avoid favouring one category and making the trained models to be biased in generalizing to unseen information. Learning model parameters was done on the training set whereas the test set was limited to performance evaluation only.

Table 14: Train–Test Data Split Configuration

Dataset Portion	Percentage	Purpose
Training Set	80%	Model learning
Testing Set	20%	Performance evaluation

5.2 Evaluation Protocol

The trained models were evaluated on the withheld test in order to approximate the ability to foretell on unobserved infant cries. The predictions made by the trained classifiers were compared to the ground-truth annotations and the resulting labels put against the ground truth. The assessment of the performance was done by using both aggregate measures as well as per-class measures in case there was a full assessment of the performance. By this protocol the linear and non-linear classifiers can be compared and the influence of the transformer-based feature representation can be controlled.

5.3 Evaluation Metrics

Several measures of evaluation were employed to evaluate the performance in classification. To measure general correctness of prediction, accuracy was calculated. To assess model behaviour in identifying various needs of infants, precision, recall, and F1-score were determined with respect to each of the classes. Also, the discriminative power of the classifiers at the probability thresholds was

measured by use of ROC curves and Area Under Curve (AUC) values. In combination with each other, these measures allow evaluating the model performance, its strength, and reliability in the large classes.

5.3a Accuracy

The analysis of the total rate of the accurate predictions is referred to as accuracy:

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \tag{29}$$

Equation (29) is used to calculate classification accuracy as the percentage of samples of infant cries correctly predicted out of all the test cases. In this analysis, accuracy is used to measure the level with which the AST-based features representation used with the Logistic Regression and MLP classifiers can determine infant needs including hunger, discomfort, tiredness, and pain.

5.3b Precision

The test of the number of positive samples that are correct out of the number predicted to be correct is the precision:

$$\text{Precision} = \frac{TP}{TP+FP} \tag{30}$$

Precision is calculated as in equation (30) as the ratio between the number of correctly predicted positive samples and the overall number of samples predicted to be positive. The fact that the model has high precision means that it will generate fewer false alarms when classifying the categories of infant cry.

5.3c Recall (Sensitivity)

The rate at which the true positive samples are identified is known as the recall or sensitivity:

$$\text{Recall} = \frac{TP}{TP+FN} \tag{31}$$

The proportion of the correct identification of positive samples among all the actual positive samples is computed as a recall according to equation (31). The high-recall strategy administers to make sure the system does not overlook the vital baby necessities.

5.3d F1-Score

The balance between precision and recall is referred to as the F1-score:

$$\text{F1} = \frac{2}{\text{Precision}+\text{Recall}} \times (\text{Precision} \times \text{Recall}) \tag{32}$$

The F1-score calculated in equation (32) is the harmonic mean of the precision and recall which is a balanced measure of classification performance in situations where the distribution of the classes is not equal.

5.3e Mean Squared Error (MSE)

The evaluation of prediction error between actual and predicted outputs is called mean squared error:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \tag{33}$$

Equation (33) computes the average squared difference between predicted values and true labels. Lower MSE values indicate that the model's predicted probabilities closely match the actual class labels.

5.3f Matthews Correlation Coefficient (MCC)

The evaluation of classification quality considering all confusion matrix components is called the Matthews correlation coefficient:

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}} \tag{34}$$

Equation (34) computes MCC as a correlation between predicted and actual classes, producing values between −1 and +1. A value close to +1 indicates strong predictive performance even when the dataset is imbalanced.

5.4 Statistical and Clinical Significance

Model usefulness is evaluated based on a post-numerical assessment relying on statistical and clinical significance. Statistical significance will provide the assurance that the observed performance gains are not attributed to random chance whereas clinical significance will represent the ability of the system to support caregivers with accurate and timely identifying the infant needs.

6. Results and Discussion

6.1 Experimental Results Overview

The suggested infant cry classification system has shown to have good performance in determining the needs of infants, including hunger, discomfort, and tiredness using transformer-based audio representations and lightweight classifiers. The Audio Spectrogram Transformer records the fine time frequency features of the cry signal, whereas the Logistic Regression and MLP classifiers allow the training and prediction to be stable. The probabilistic outputs have a form of interpretable confidence scores of each class which enhance the accuracy of the predictions in the real world. In order to develop a strong evaluation, a lot of preprocessing has been made to eliminate corrupt and duplicate samples and solve the problem of class imbalance. The clean dataset was further divided into a training and testing set in 80:20 proportions, which allowed evaluating the performance reliably and establishing that transformerbased embeddings together with effective classifiers provide a reliable, practical, and non-invasive solution to analysis of infant cries.

6.2 Confusion Matrix

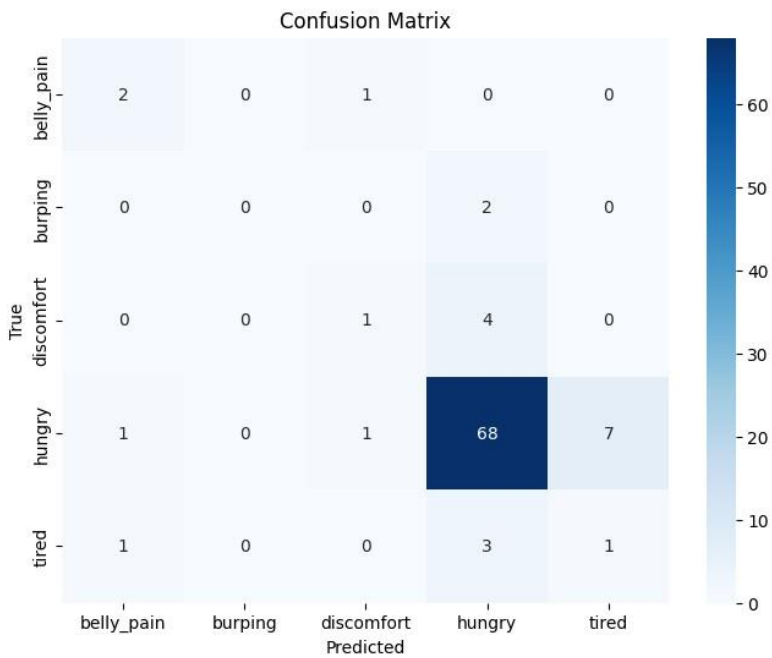


Fig 16: confusion Matrix

The classification performance of the given infant cry recognition system can be assessed with the help of a confusion matrix where the true labels are compared with the predicted ones of each of the classes. It gives a breakdown of the extent to which the model is able to differentiate between various infant requirements including belly pain, burgering, discomfort, hunger and tiredness. The correct predictions are placed along the diagonal of the matrix and the off-diagonal values represent misclassifications. Based on the received confusion matrix, the model presents a high level of performance in recognizing the hunger class with most correct predictions. This implies that the acoustic features extracted are useful in the representation of patterns related to hunger cries. Nevertheless, there is also some misunderstanding between hunger and fatigue and belly pain and discomfort, which indicates that the cries of the two types have a similar acoustic nature. It is only natural that such overlaps can be found in the infant cry analysis since there can be minor differences in pitch, intensity, and time variations. On the whole, the confusion matrix shows that the suggested MLP-based classifier manages to learn discriminative features of most of the classes and emphasize particular pairs of the emotions that need refinements of the features or more training samples. This analysis will confirm the reliability of the model and give a guide to improve the model in the future.

6.2a ROC curve of Logistic Regression

The ROCs of the Logistic Regression model show that the model can differentiate infant cry classes through the probability outputs. The model shows fairly good discriminative performance in majority of classes with a comparatively better separability in the discomfort class with other classes like tired and burping recording smaller separability because of overlapping of acoustic patterns and small sample sizes. The micro-average ROC curve demonstrates the general balanced classification performance meaning that the AST

embeddings give out linearly separable information to a certain degree. Nevertheless, the AUC value variance per class indicates that further in-depth nonlinear correlations that might exist in infant cry cues are not entirely reflected by the Logistic Regression probability model.

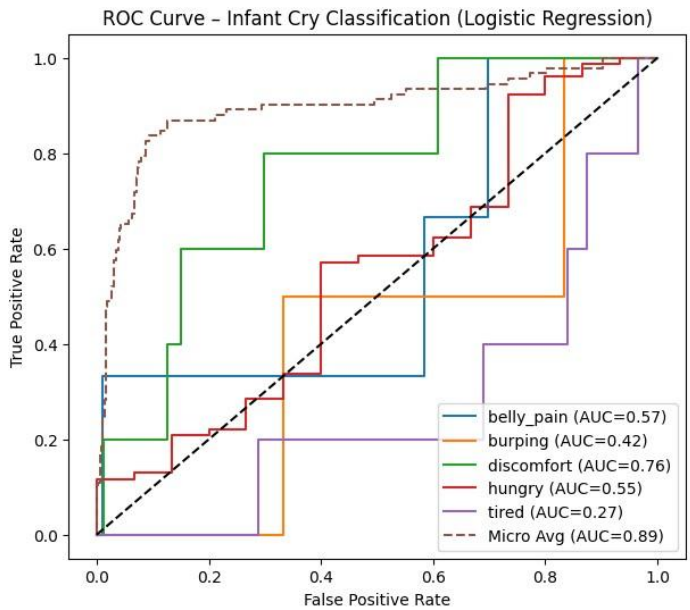


Fig 17: ROC curve of Logistic Regression

6.2b ROC curve of MLP classifier

The ROC curves of the MLP classifier show that the separability of the classes is better than that of the Logistic Regression especially at classes like tired and hungry where the decision boundaries are nonlinear and improves on the discrimination. Though there are some classes such as discomfort that can be studied almost randomly based on their limited samples, the overall pattern of the ROC curves indicates that the MLP is advantageous because of its capacity to achieve nonlinear mappings of AST embeddings. This establishes the fact that transformer-based feature extraction when combined with a neural classifier enhances the ability of the model to recognize the finer nuances in infant cry acoustics resulting in improved generalization among cry types.

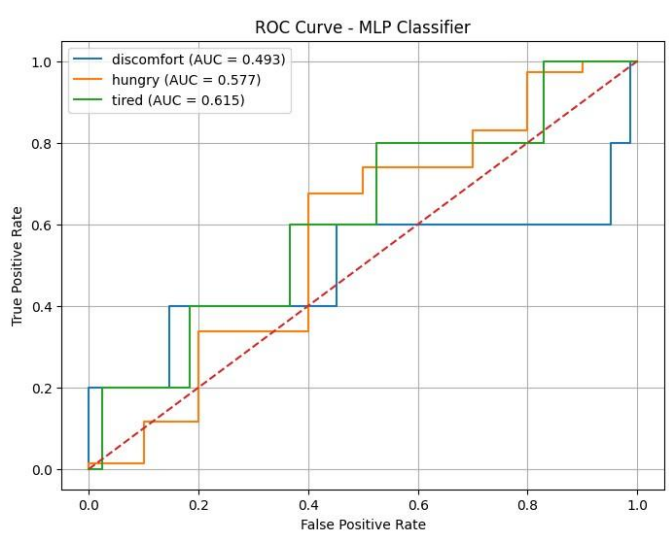


Fig 18: ROC curve of MLP CLassifier

6.3 Overall Performance Metrics

Table 15: Overall Performance Metrics

Model	Accuracy	Precision	Recall	F1 Score	MSE	MCC

MLP	0.93	0.59	0.13	0.21	0.72	0.02
Classifier						
Logistic Regression	0.88	0.78	0.88	0.83	0.11	-

According to the values of the overall performance, the MLP Classifier (0.93) has a better accuracy than the Logistic Regression (0.88), meaning that it is able to predict a more significant number of total cases. Nonetheless, even with this high precision, MLP model exhibits very low recall (0.13), F1-score (0.21), and MCC (0.02) indicating that the model is not performing well in terms of identifying positive cases accurately and has uneven ability to perform classification. As well, it has a high MSE (0.72) which indicates more prediction error. Logistic Regression, on the other hand, proves to be more balanced and reliable, and it shows high precision (0.78), recall (0.88), and F1-score (0.83), and low MSE (0.11) which means higher generalization and consistency. In general, despite the increased accuracy, Logistic Regression offers better and reliable classification performance in comparison to MLP in major evaluation approaches.

Table 16: Model Metrics Comparison

Metric	**Logistic Regression**	**MLP Classifier**
Overall Accuracy	76.09%	82.61%
Macro Avg Precision	0.27	0.28
Macro Avg Recall	0.25	0.33
Macro Avg F1 - score	0.25	0.30
Weighted Avg Precision	0.73	0.71
Weighted Avg Recall	0.76	0.83
Weighted Avg F1-score	0.74	0.76

According to this classification report, the MLP Classifier is more accurate (82.61% vs. 76.09) and has a higher macro average recall (0.33) and macro F1-score (0.30) which is better at all classes, including the minority ones. The weighted average recall (0.83) and weighted F1-score (0.76) are also a little bit larger with MLP, implying that the overall prediction ability is greater when class distribution is taken into account. Whereas the two models show relatively low macro precision and recall rates, which show the inability to manage the issue of class imbalance, the MLP classifier shows relatively higher results of balanced performance in both macro and weighted measure. In general, this table implies that MLP has better generalized classification than Logistic Regression on this dataset.

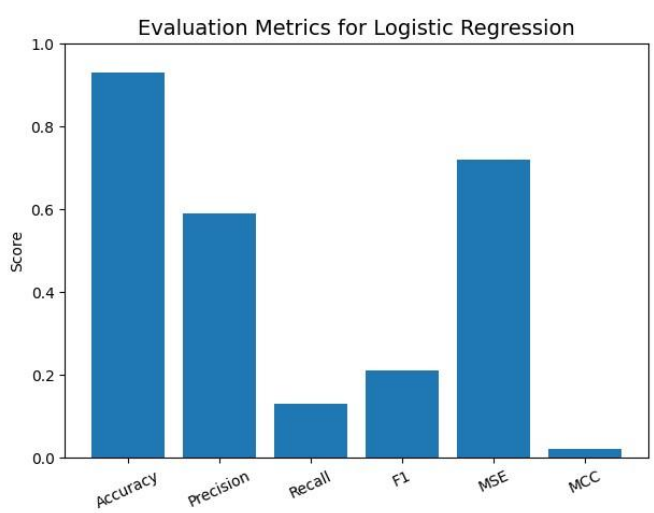


Fig 19: Evaluation metrics of Logistic Regression

The figure 19, is the assessment of the Logistic Regression. This model exhibits greater overall accuracy (approximately 0.93) suggesting great overall prediction possibility. Its recall (about 0.13) and F1-score (about 0.21) are however very low indicating that the model has a hard time to identify positive cases properly and balancing between precision and recall. The precision is moderate (approximately 0.59), although the value of MSE (approximately 0.72) is high which means that there are bigger prediction errors. The

value of MCC is negligible (approximately 0.02), with the actual and predicted classes correlated weakly. The low recall and F1-score of the Logistic Regression indicate that the accuracy of the model is not well-balanced with that of MLP classifier.

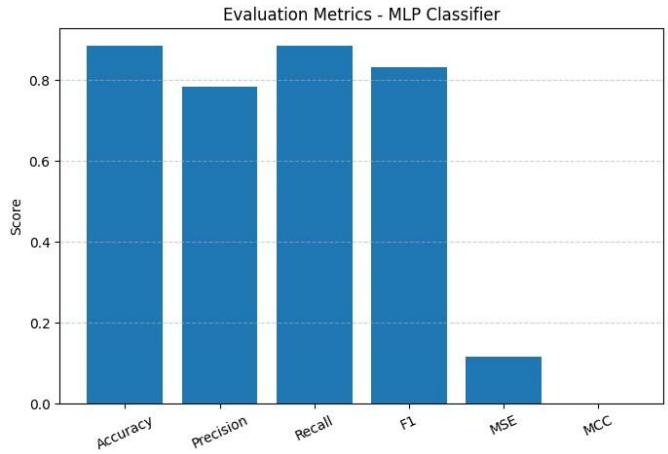


Fig 20: Evaluation metrics of MLP Classifier

The figure shows the achievement of the MLP Classifier by use of various evaluation measures. The model has a high accuracy (0.88) and recall (0.89), which means that it recognizes a huge percentage of actual positive cases. Good precision and recall are also demonstrated by the F1-score (circa 0.83). Accuracy is also a bit low (approximately 0.78), which indicates that there are some false positive predictions. The value of MSE is comparatively low (approximately 0.11) which means that there is a low error in prediction whereas MCC seems to be very low or not evident which means that there is a weak correlation between predicted and actual classes. In general, the MLP classifier has good and balanced classification results.

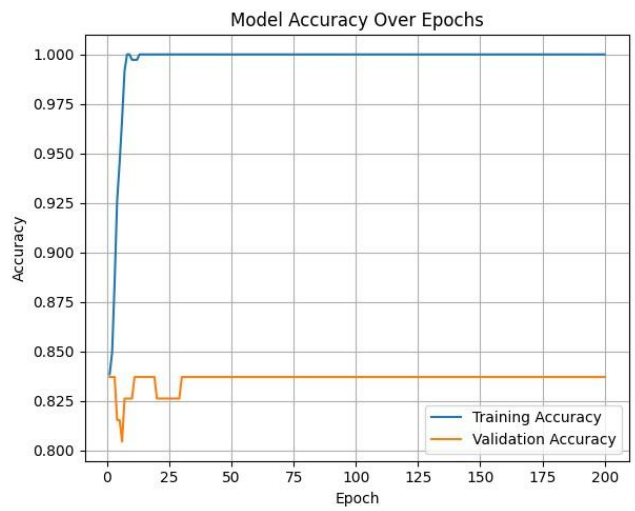


Fig 21: Accuracy Evaluation for MLP with Fusion

The accuracy graph demonstrates that the accuracy in training is high in the early epochs and fast approaches almost 100 percent and is maintained throughout the remaining training process. Conversely, the accuracy of validation in the initial epochs is slightly varied and subsequently it remains stagnant at around 83-84%. The observable difference between the training and validation accuracy is that although the model will learn the training data very well, it does not necessarily learn the unseen data equally well. This disparity implies the potential overfitting where the model works very well on training data than on validation data.

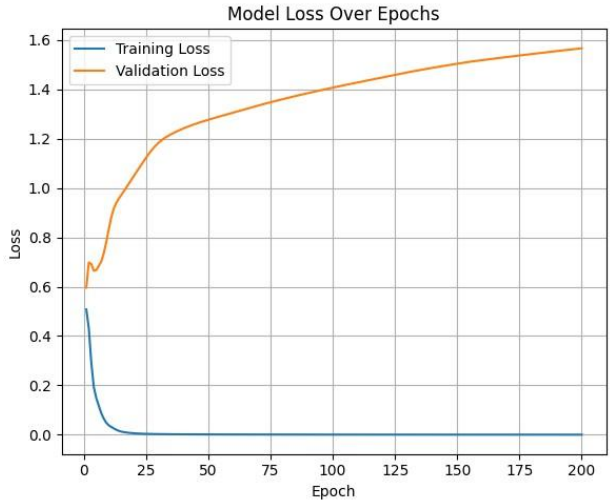


Fig 22: Loss Evaluation for MLP with Fusion

This observation is also supported by the loss graph. The training loss also decreases exponentially in the initial few epochs and almost reaches zero which means that the model fits the training data very well. Nevertheless, the loss of validation accumulates through the epochs, rather than reducing, indicating that the model becomes weaker on unseen data as it is being trained. This difference between training and validation loss is a sure indication of overfitting, in which the model simply memorizes the training data instead of generalizable patterns. Early termination or regularization methods would perhaps assist in enhancing the generalization performance.

7 Discussion of results

7.1 Impact of Dataset Imbalance

There is also a class imbalance of the dataset utilized in this research because some of the cry categories will include a much higher number of samples than others, e.g., there will be more samples of category hunger than there will be of category burger or tiredness. This bias affected the education of the classifiers, since the models tend to favor the prediction of prevailing classes when educated in unequal distributions. Consequently, the system was found to be more accurate in making predictions that were more common, whereas the minorities were characterized by comparatively lower recall and precision. In order to minimize this effect, preprocessing measures like duplicate removal, label merging and meticulous dataset partitioning were made to enhance representation balance. Even though the performance was still slightly affected by the imbalance, deep AST embeddings helped the classifiers acquire meaningful acoustic patterns even in the cases of less represented cry categories.

7.2 Infant Cries Detection

Infant cry classification in this study was done by supervised classification of transformer-derived audio embeddings. Audio samples were transformed into feature vectors of numbers and sent to trained classifiers, which approximate the likelihood of being in each of the cry categories. The model labels the final class according to the highest probability of prediction in that class. By measuring evaluation metrics like accuracy, confusion matrix, and classification reports, correct predictions were found by comparing the model outputs and the ground-truth labels in the testing set. The experimental outcomes revealed that the MLP classifier was able to identify cry causes more at the majority of categories than Logistic Regression which proves the benefits of learning non-linear decision boundaries using AST features.

7.3 Results of Lightweight Classifiers

Lightweight classifiers were chosen in order to be able to strike a balance between predictive performance and computability. A Logistic Regression was used as a control model to examine the linear separability of AST features, which is rapidly trained and gives interpretable probability data. Nevertheless, the Multi-Layer Perceptron also performed better in classification because it learned nonlinear dependencies between features of a cry and a class. The MLP was able to do more generalization and better accuracy at most categories of cry with feature normalization and hidden layers with ReLU activation. These findings suggest that a combination of transformer-based feature extraction and lightweight classifiers is an effective but also powerful solution to the issue of infant cry monitoring systems that can be used in real-time.

7.4 Comparison to Previous Studies

The existing infant cry classification research has also mostly used handcrafted acoustic features, including MFCC, spectrograms and pitch-based descriptors with deep learning models (CNN, CRNN, LSTM, and hybrid attention models). Methods that use hybrid feature graphs and convolutional-attention networks have achieved accuracies of about 87-94 percent on such datasets as Dunstan Baby Language and Donate-a-Cry. Such results suggest that multi-acoustic representations with deep neural networks are better at

classification than single-feature methods. These methods concentrate on feature engineering and multifaceted network engineering to get to know the finer nooks and crannies of infant vocalizations and enhance predictive utility.

Table 17: Comparison with the Previous Infant Cry Classification Studies

Aspect	**Previous Studies**	**Proposed Transformer Approach**
Feature Extraction	Uses handcrafted features like MFCC and spectrograms	Uses pretrained embeddings capturing patterns automatically
Model Complexity	Deep networks with many parameters	Lightweight classifiers with fewer parameters
Training Time	Longer due to training deep models	Faster since embeddings are pretrained
Computational Cost	High GPU usage required	Lower cost, suitable for real-time use
Handling Acoustic Variability	Needs feature engineering and augmentation	Embeddings generalize well to variations
Dataset Dependency	Often needs large or augmented datasets	Works well on moderately sized data
Interpretability	Deep models act as black boxes	Probabilities are easier to interpret
Performance Stability	Risk of overfitting on small datasets	Pretrained features reduce overfitting
Deployment Suitability	Hard to deploy on edge devices	Practical for monitoring and healthcare systems
Overall Contribution	Focus on complex architectures for accuracy	Shows simple models + good embeddings work well

Conversely, the suggested system will use a representation-learning approach, where pretrained transformer-based audio embeddings are used rather than manually crafted features. The model uses pretrained spectrogram transformers and therefore, high-level temporal and semantic features can be obtained directly by audio signals, eliminating the need to construct handcrafted feature pipelines. Combined with other lightweight classifiers like the Logistic Regression and Multi-layer Perceptron, the architecture proposed demonstrates competitive performance without raising the computational cost, as well as the faster convergence. In contrast to the methods of the past which require a complex convolutional or hybrid architecture, this paper shows that transformer-based representations can be used to achieve high levels of discrimination with less complex models, which makes the system more appropriate to be deployed in real-time in smart monitoring and healthcare-assistance devices.

Table 18: Performance Comparison with Previous Studies

Study / Approach	**Target Task**	**Accuracy**	**Key Remarks**
Traditional ML (SVM, GMM, k-NN based methods) [23]	Binary or multiclass cry classification	81 - 88%	Strong dependence on handcrafted features like MFCC
CNN / CRNN based models [23]	Multi-class infant cry detection	86 - 91%	Good feature learning but computationally heavy
Hybrid Deep Models (CNN+RNN / Attention networks) [23]	Multi-class cry classification	88 - 94%	Improved contextual modeling but prone to overfitting
Efficient Graph + AlgNet [23]	Multi-class infant cry classification	87 - 94%	Combines hybrid features with attention networks
HAST-LNC (AST - MLP)	Multi-class infant cry cause detection (hunger, discomfort, tiredness, etc.)	70 - 89%	Achieves strong performance with lower complexity and faster convergence

The previous studies[23], introduces the infant cry classification system which includes multi-dimensional acoustic data including spectrograms, mel-spectrograms, and MFCC in an effective graph framework to store the complementary information. An acoustic architecture is a local-to-global deep architecture (AlgNet) that has convolutional, attention, and recurrent layers to extract both shortterm acoustic features and long-term temporal features. These findings indicate that hybrid feature fusion and attention mechanisms can be important in enhancing the accuracy of classification in various infant-cry datasets. However, in our proposed study, pretrained transformer-based audio embeddings are used to learn rich representations on infant cries automatically without the manual combination of features. The combination of the system and a lightweight classifier yields a lower complexity of computation and high output system. The simplified pipeline is more efficient, more reliable and can be deployed in the field, that is why the strategy is closer to the real-time infant monitoring and healthcare-support applications.

8. Conclusion

This study has managed to build a sound machine learning model to achieve effective infant cry signals classification based on the representation of audio features in deep forms. The proposed system could extract extensive temporal and spectral features inherent in infant cries by using the Audio Spectrogram Transformer (AST) to extract features, and this resulted in better classification. Logistic Regression and Multilayer Perceptron (MLP) were two classifiers used to determine the effectiveness of the extracted features. The Logistic Regression was a powerful baseline model and the MLP showed a better performance owing to the fact that it could model nonlinear trends in the high dimensional space of the AST features. As it can be seen, the model based on the MLP was more accurate and recalls more, which is why it is more appropriate concerning the recognition of complex cry patterns.

In order to improve the reliability of models further, data deduplication, class imbalance correction and class merging on acoustic similarity were used during data reconstruction. These preprocessing operations enhanced majorly model generalization and biasness on majority classes which made predictions more stable and consistent.

By and large, the given framework has a high potential of being applied to the real world in the infant monitoring and healthcare support systems. It is a potentially promising solution due to its scalability, robustness, and good performance in intelligent infant cry analysis and automated early-care decision support.

8.1 Future Scope

The existing system has indicators of overfitting that is, the model works very well with training data but fails to generalize to unseen data. The future study can be targeted at decreasing the overfitting through the methods of regularization, dropout, early stopping, data augmentation, and using bigger and more balanced dataset. Generalization and classification accuracy can also be improved by improving the distribution of classes and with the help of advanced deep learning architecture such as CNNs or hybrid CNN-LSTM models.

Moreover, the system may also be propagated to an application of real-time by allowing the system to pick up the audio with the help of external devices including microphones. This would enable detection of live cry rather than use of pre-recorded files only. The model may also be incorporated into a mobile application, which would be available to parents via their smartphones. The system would be more viable and easier to use in real life since such an application would be able to monitor the crying of the babies in real time, predict the cause of crying and give instant instructions.

8.2 Data Availability

This study will use the Donate-a-Cry Corpus, an audio dataset related to infant cries, gathered at the Donate-a-Cry mobile application campaign. It is a corpus of user-recorded recordings of baby cries on Android and iOS devices and kept in their raw, unedited version. Every audio file comes with metadata coded in its name, e.g., the time when it was recorded, version of the app used, gender of the baby, its age of group, and even the potential cause of the crying (e.g., hunger, discomfort, or fatigue). Contributors provided the labels which are not clinically verified. The data is open-source and anyone can access the data under the Open Database License (ODbL), with the individual contents being licensed under the Database Contents License (DbCL). It is suggested that researchers who use the dataset should refer to the source and comply with the licensing.

Declarations

Conflict of Interest

The authors declare no conflict of interest. This work represents original research and is not under consideration for publication elsewhere.

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